BSDS FINAL PROJECT

Gamefide

A Gamified STEM and English Learning Application for Children

Advisor: <Advisor Name>

Table 1: Group members placeholder table

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Abstract

In the modern educational landscape, traditional learning methods are increasingly being replaced or enhanced by interactive, technology-driven solutions that improve engagement, motivation, and knowledge retention among students. The Gamefide project is a comprehensive gamified learning platform designed specifically for children aged 7 to 12 years, aiming to transform conventional STEM (Science, Technology, Engineering, and Mathematics) and English education into an immersive, game-based learning experience.

The primary objective of Gamefide is to bridge the gap between education and entertainment by integrating structured curriculum content with interactive mini-games, quizzes, and AI-powered storytelling. The system is composed of two major components: a Child Portal and a Parent Portal. The Child Portal provides an engaging environment where students can participate in subject-based games, complete progressive quizzes, earn points, maintain learning streaks, and unlock achievements. In contrast, the Parent Portal enables guardians to monitor their child's academic progress, view detailed performance analytics, manage profiles, and track learning behavior through graphical reports and dashboards.

The system is developed using a modern full-stack architecture. The frontend is built with Ionic Framework and Angular, providing a responsive cross-platform user interface compatible with web and mobile devices. The backend is implemented using Node.js and Express.js, ensuring scalable API development and efficient request handling. Data persistence is managed through MongoDB, which supports flexible schema design for storing user profiles, game progress, quizzes, achievements, and AI-generated content. Additionally, Socket.io is used to support real-time updates, while JWT-based authentication ensures secure access control for both parents and children.

A key innovation of the Gamefide platform is the integration of Google Gemini AI (gemini-2.5-flash model), which generates personalized interactive stories based on child inputs such as character names, selected animals, and learning themes. This feature enhances creativity and reading engagement while reinforcing moral and educational lessons. Furthermore, structured quizzes and level-based challenges ensure progressive learning aligned with STEM educational goals.

The system includes multiple categorized educational mini-games across Mathematics, Science, Technology, and English domains. Each game is carefully designed to reinforce specific cognitive and academic skills such as problem-solving, pattern recognition, language development, and logical reasoning. Progress tracking is implemented through a scoring system, learning streaks, and achievement badges, which encourage continuous participation and healthy competition among learners.

In conclusion, Gamefide presents a scalable, interactive, and intelligent learning ecosystem that combines educational content, gamification principles, and artificial intelligence to enhance early childhood education. The platform not only supports students in learning complex concepts in an engaging manner but also empowers parents to actively participate in their child's educational journey through real-time insights and analytics.

Dedication

We dedicate this project to Almighty Allah (SWT), the Most Merciful and the Most Gracious, who blessed us with the strength, knowledge, patience, and guidance throughout the completion of this Final Year Project. Without His countless blessings, the successful completion of this work would not have been possible.

We would also like to dedicate this project to our respected parents and families, whose unconditional love, continuous support, encouragement, and prayers have been the foundation of our success. Their belief in our abilities and constant motivation kept us going through every challenge we faced during our academic journey. This achievement is as much theirs as it is ours.

We extend our sincere gratitude to our honorable project supervisor and faculty members of the Faculty of Information Technology, University of Okara, for their valuable guidance, insightful feedback, and continuous encouragement throughout this project. Their expertise and mentorship helped us shape this project into a complete and functional system.

We also dedicate this work to all teachers who have played a role in shaping our academic careers. Their dedication towards education and knowledge sharing has inspired us to pursue excellence in the field of Computer Science and software development.

Furthermore, we dedicate this project to future students and learners who will use or improve upon this work. We hope that the Gamefide platform serves as a helpful contribution to the field of gamified learning systems and inspires further innovation in educational technology.

Finally, this project is dedicated to all those who believe in the power of education, creativity, and technology to transform learning into an engaging and meaningful experience.

Acknowledgements

First and foremost, we would like to express our deepest gratitude to Almighty Allah (SWT), whose blessings, mercy, and guidance enabled us to complete this Final Year Project successfully. It is only through His infinite wisdom and strength that we were able to overcome all challenges during this journey.

We are highly grateful to our respected project supervisor, [Advisor Name], for their continuous guidance, valuable suggestions, and constructive feedback throughout the development of this project. Their support and encouragement played a vital role in shaping the direction and quality of our work. Their experience in the field of software engineering helped us refine both our technical and analytical skills.

We would also like to extend our sincere thanks to all faculty members of the Faculty of Information Technology, University of Okara, for their cooperation, encouragement, and knowledge-sharing throughout our academic journey. Their teaching and mentorship provided us with a strong foundation in computer science and software development.

Special thanks go to our parents and families, whose unconditional love, prayers, and moral support have always been a source of strength and motivation. Their encouragement kept us focused and determined even during the most challenging phases of this project.

We are also thankful to our friends and classmates for their continuous support, collaboration, and helpful discussions during the development of this project. Their feedback and ideas helped us improve our system in multiple ways.

Finally, we acknowledge all those who directly or indirectly contributed to the successful completion of this project. This work would not have been possible without their support, guidance, and encouragement.

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Revision History

Table 2: Revision history

Name	Date	Reason for Changes	Version
Project Team	Initial Phase	Proposal and concept	0.1
Project Team	Design Phase	Architecture, database, diagrams, prototype	0.5
Project Team	Implementation Phase	Frontend, backend, games, AI, reports	0.9
Project Team	Final Documentation	Complete final report with 80+ pages	1.0

Chapter 1. Technologies Used and Technical Foundations

Gamefide is a complete full-stack system. This chapter is placed first because the project uses several technologies across frontend, backend, database, native integration, charting, security, API documentation, real-time capability, and generative AI. The technologies were selected to make the system mobile-ready, maintainable, secure, and extensible.

Table 3: Technology stack overview

Technology	Use in Gamefide	Why selected
Ionic Framework	Builds mobile-like pages, tabs, cards, modals, and child-friendly screens.	Suitable for a modular educational application and aligned with project requirements.
Angular 20	Organizes the frontend into standalone components, services, guards, routing, and forms.	Suitable for a modular educational application and aligned with project requirements.
Capacitor 7	Packages the web application for Android/iOS and supports native plugins.	Suitable for a modular educational application and aligned with project requirements.
TailwindCSS and SCSS	Provide rapid styling, responsive layouts, gradients, and custom themes.	Suitable for a modular educational application and aligned with project requirements.
ApexCharts	Visualizes parent reports, learning trends, scores, and progress summaries.	Suitable for a modular educational application and aligned with project requirements.
Native biometric plugin	Supports fingerprint-based child login on compatible devices.	Suitable for a modular educational application and aligned with project requirements.
Capacitor social login	Allows parent login through Google/Apple identity flows.	Suitable for a modular educational application and aligned with project requirements.
Node.js	Runs the backend server and handles REST requests efficiently.	Suitable for a modular educational application and aligned with project requirements.
Express.js	Defines API routes and middleware for authentication, games, quizzes, reports, and stories.	Suitable for a modular educational application and aligned with project requirements.
MongoDB	Stores flexible child profiles, achievements, nested quiz data, and stories.	Suitable for a modular educational application and aligned with project requirements.
Mongoose ODM	Defines schemas, references, validation, defaults, indexes, and hooks.	Suitable for a modular educational application and aligned with project requirements.
JWT	Protects parent, child, and admin routes using token-based authentication.	Suitable for a modular educational application and aligned with project requirements.
Joi	Validates incoming request payloads before controller execution.	Suitable for a modular educational application and aligned with project requirements.
Helmet and CORS	Improve HTTP security headers and control frontend access to backend APIs.	Suitable for a modular educational application and aligned with project requirements.
Socket.io	Provides a foundation for future real-time feedback and notifications.	Suitable for a modular educational application and aligned with project requirements.
Swagger UI	Documents REST endpoints for testing and review.	Suitable for a modular educational application and aligned with project requirements.
Google Gemini API	Generates personalized bedtime, math, moral, and adventure stories.	Suitable for a modular educational application and aligned with project requirements.

1.1 Ionic Framework

Ionic Framework is used in Gamefide to builds mobile-like pages, tabs, cards, modals, and child-friendly screens.. It contributes to the final product by improving development speed, reliability, maintainability, or user experience. In the implemented system, this technology works with other layers rather than operating alone. For example, frontend components call Angular services, services call Express routes, routes use middleware and Mongoose models, and the database stores the results.

The project team selected Ionic Framework because it supports the educational product goals: a responsive child interface, secure parent access, flexible data storage, useful reports, and future scalability. The technology also provides a professional implementation foundation for a university final year project.

1.2 Angular 20

Angular 20 is used in Gamefide to organizes the frontend into standalone components, services, guards, routing, and forms.. It contributes to the final product by improving development speed, reliability, maintainability, or user experience. In the implemented system, this technology works with other layers rather than operating alone. For example, frontend components call Angular services, services call Express routes, routes use middleware and Mongoose models, and the database stores the results.

The project team selected Angular 20 because it supports the educational product goals: a responsive child interface, secure parent access, flexible data storage, useful reports, and future scalability. The technology also provides a professional

implementation foundation for a university final year project.

1.3 Capacitor 7

Capacitor 7 is used in Gamefide to packages the web application for Android/iOS and supports native plugins.. It contributes to the final product by improving development speed, reliability, maintainability, or user experience. In the implemented system, this technology works with other layers rather than operating alone. For example, frontend components call Angular services, services call Express routes, routes use middleware and Mongoose models, and the database stores the results.

The project team selected Capacitor 7 because it supports the educational product goals: a responsive child interface, secure parent access, flexible data storage, useful reports, and future scalability. The technology also provides a professional implementation foundation for a university final year project.

1.4 TailwindCSS and SCSS

TailwindCSS and SCSS is used in Gamefide to provide rapid styling, responsive layouts, gradients, and custom themes.. It contributes to the final product by improving development speed, reliability, maintainability, or user experience. In the implemented system, this technology works with other layers rather than operating alone. For example, frontend components call Angular services, services call Express routes, routes use middleware and Mongoose models, and the database stores the results.

The project team selected TailwindCSS and SCSS because it supports the educational product goals: a responsive child interface, secure parent access, flexible data storage, useful reports, and future scalability. The technology also provides a professional implementation foundation for a university final year project.

1.5 ApexCharts

ApexCharts is used in Gamefide to visualizes parent reports, learning trends, scores, and progress summaries.. It contributes to the final product by improving development speed, reliability, maintainability, or user experience. In the implemented system, this technology works with other layers rather than operating alone. For example, frontend components call Angular services, services call Express routes, routes use middleware and Mongoose models, and the database stores the results.

The project team selected ApexCharts because it supports the educational product goals: a responsive child interface, secure parent access, flexible data storage, useful reports, and future scalability. The technology also provides a professional implementation foundation for a university final year project.

1.6 Native biometric plugin

Native biometric plugin is used in Gamefide to supports fingerprint-based child login on compatible devices.. It contributes to the final product by improving development speed, reliability, maintainability, or user experience. In the implemented system, this technology works with other layers rather than operating alone. For example, frontend components call Angular services, services call Express routes, routes use middleware and Mongoose models, and the database stores the results.

The project team selected Native biometric plugin because it supports the educational product goals: a responsive child interface, secure parent access, flexible data storage, useful reports, and future scalability. The technology also provides a professional implementation foundation for a university final year project.

1.7 Capacitor social login

Capacitor social login is used in Gamefide to allows parent login through Google/Apple identity flows.. It contributes to the final product by improving development speed, reliability, maintainability, or user experience. In the implemented system, this technology works with other layers rather than operating alone. For example, frontend components call Angular services, services call Express routes, routes use middleware and Mongoose models, and the database stores the results.

The project team selected Capacitor social login because it supports the educational product goals: a responsive child interface, secure parent access, flexible data storage, useful reports, and future scalability. The technology also provides a professional implementation foundation for a university final year project.

1.8 Node.js

Node.js is used in Gamefide to runs the backend server and handles REST requests efficiently.. It contributes to the final product by improving development speed, reliability, maintainability, or user experience. In the implemented system, this technology works with other layers rather than operating alone. For example, frontend components call Angular services, services call Express routes, routes use middleware and Mongoose models, and the database stores the results.

The project team selected Node.js because it supports the educational product goals: a responsive child interface, secure parent access, flexible data storage, useful reports, and future scalability. The technology also provides a professional implementation foundation for a university final year project.

1.9 Express.js

Express.js is used in Gamefide to defines API routes and middleware for authentication, games, quizzes, reports, and stories.. It contributes to the final product by improving development speed, reliability, maintainability, or user experience. In the implemented system, this technology works with other layers rather than operating alone. For example, frontend components call Angular services, services call Express routes, routes use middleware and Mongoose models, and the database stores the results.

The project team selected Express.js because it supports the educational product goals: a responsive child interface, secure parent access, flexible data storage, useful reports, and future scalability. The technology also provides a professional implementation foundation for a university final year project.

1.10 MongoDB

MongoDB is used in Gamefide to stores flexible child profiles, achievements, nested quiz data, and stories.. It contributes to the final product by improving development speed, reliability, maintainability, or user experience. In the implemented system, this technology works with other layers rather than operating alone. For example, frontend components call Angular services, services call Express routes, routes use middleware and Mongoose models, and the database stores the results.

The project team selected MongoDB because it supports the educational product goals: a responsive child interface, secure parent access, flexible data storage, useful reports, and future scalability. The technology also provides a professional implementation foundation for a university final year project.

1.11 Mongoose ODM

Mongoose ODM is used in Gamefide to defines schemas, references, validation, defaults, indexes, and hooks.. It contributes to the final product by improving development speed, reliability, maintainability, or user experience. In the implemented system, this technology works with other layers rather than operating alone. For example, frontend components call Angular services, services call Express routes, routes use middleware and Mongoose models, and the database stores the results.

The project team selected Mongoose ODM because it supports the educational product goals: a responsive child interface, secure parent access, flexible data storage, useful reports, and future scalability. The technology also provides a professional implementation foundation for a university final year project.

1.12 JWT

JWT is used in Gamefide to protects parent, child, and admin routes using token-based authentication.. It contributes to the final product by improving development speed, reliability, maintainability, or user experience. In the implemented system, this technology works with other layers rather than operating alone. For example, frontend components call Angular services, services call Express routes, routes use middleware and Mongoose models, and the database stores the results.

The project team selected JWT because it supports the educational product goals: a responsive child interface, secure parent access, flexible data storage, useful reports, and future scalability. The technology also provides a professional implementation foundation for a university final year project.

1.13 Joi

Joi is used in Gamefide to validates incoming request payloads before controller execution.. It contributes to the final product by improving development speed, reliability, maintainability, or user experience. In the implemented system, this technology works with other layers rather than operating alone. For example, frontend components call Angular services, services call Express routes, routes use middleware and Mongoose models, and the database stores the results.

The project team selected Joi because it supports the educational product goals: a responsive child interface, secure parent access, flexible data storage, useful reports, and future scalability. The technology also provides a professional implementation foundation for a university final year project.

1.14 Helmet and CORS

Helmet and CORS is used in Gamefide to improve HTTP security headers and control frontend access to backend APIs.. It contributes to the final product by improving development speed, reliability, maintainability, or user experience. In the implemented system, this technology works with other layers rather than operating alone. For example, frontend components call Angular services, services call Express routes, routes use middleware and Mongoose models, and the database stores the results.

The project team selected Helmet and CORS because it supports the educational product goals: a responsive child interface, secure parent access, flexible data storage, useful reports, and future scalability. The technology also provides a professional implementation foundation for a university final year project.

1.15 Socket.io

Socket.io is used in Gamefide to provides a foundation for future real-time feedback and notifications.. It contributes to the final product by improving development speed, reliability, maintainability, or user experience. In the implemented system, this technology works with other layers rather than operating alone. For example, frontend components call Angular services, services call Express routes, routes use middleware and Mongoose models, and the database stores the results.

The project team selected Socket.io because it supports the educational product goals: a responsive child interface, secure parent access, flexible data storage, useful reports, and future scalability. The technology also provides a professional implementation foundation for a university final year project.

1.16 Swagger UI

Swagger UI is used in Gamefide to documents REST endpoints for testing and review.. It contributes to the final product by improving development speed, reliability, maintainability, or user experience. In the implemented system, this technology works with other layers rather than operating alone. For example, frontend components call Angular services, services call Express routes, routes use middleware and Mongoose models, and the database stores the results.

The project team selected Swagger UI because it supports the educational product goals: a responsive child interface, secure parent access, flexible data storage, useful reports, and future scalability. The technology also provides a professional implementation foundation for a university final year project.

1.17 Google Gemini API

Google Gemini API is used in Gamefide to generates personalized bedtime, math, moral, and adventure stories.. It contributes to the final product by improving development speed, reliability, maintainability, or user experience. In the implemented system, this technology works with other layers rather than operating alone. For example, frontend components call Angular services, services call Express routes, routes use middleware and Mongoose models, and the database stores the results.

The project team selected Google Gemini API because it supports the educational product goals: a responsive child interface, secure parent access, flexible data storage, useful reports, and future scalability. The technology also provides a professional implementation foundation for a university final year project.

Table 4: Technology evaluation matrix

Decision Area	Selected	Alternative	Justification
Frontend	Ionic + Angular	Flutter / React Native	One codebase for browser and mobile with Angular services.
Backend	Node.js + Express	Django / Spring Boot	Lightweight REST API development and JavaScript ecosystem.
Database	MongoDB + Mongoose	MySQL / PostgreSQL	Flexible nested game, quiz, story, and achievement data.
AI	Google Gemini API	Static stories	Personalized story generation with backend prompt control.
Charts	ApexCharts	Plain tables	Clear visual reports for parents.

Chapter 2. Introduction

2.1 Product Problem Statement

Gamefide solves the problem of converting children's screen time into structured learning time. It combines games, quizzes, progress, achievements, parent monitoring, and AI stories in one application.

2.2 Background

Educational applications often focus on a single subject or activity type. Gamefide combines gamification, parental visibility, subject-wise games, and AI-generated reading content.

2.3 Objectives

The objectives are to build a dual portal system, implement child profiles and login, create educational games, track progress, generate reports, integrate AI stories, and document the product completely.

2.4 Scope

The scope includes frontend screens, backend APIs, MongoDB models, authentication, child management, games, quiz progression, achievements, reports, AI storybook, testing, deployment guide, and user manual.

2.5 Business Goals

The product can be extended for home learning, schools, tuition centers, and parent-supervised digital learning. Business value comes from child engagement and parent trust.

2.6 Challenges

The team handled role separation, child-friendly UI, multiple games, progress tracking, secure authentication, AI prompts, and complete final documentation.

2.7 Learning Outcomes

The project improved skills in full-stack development, UI/UX, database design, API documentation, testing, diagrams, gamification, and generative AI integration.

2.8 Nature of End Product

The final output is a working educational software package consisting of an Ionic Angular client and Node.js Express API with MongoDB persistence.

2.9 Related Work

Compared with quiz-only or game-only applications, Gamefide provides multiple subjects, parent reports, achievements, 40-level quiz progression, and AI story generation.

2.10 Document Conventions

Chapters, subsections, tables, figures, and appendices follow the university project documentation pattern. Placeholder fields can be replaced before submission.

Table 5: Related work comparison

Aspect	Typical Quiz App	Typical Game App	Gamefide
Learning model	Questions only	Play focused	Games, quizzes, stories, achievements
Parent visibility	Basic/none	Absent	Dashboard and reports
Personalization	Fixed content	Fixed levels	AI stories and profiles
Subjects	One subject	One theme	STEM and English

Chapter 3. Overall Description

Table 6: Product features

Feature	Description	Primary User
Dual parent and child portals	Implemented in the Gamefide client and supported by backend routes, models, or services.	Parent / Child / Admin
Parent registration and dashboard	Implemented in the Gamefide client and supported by backend routes, models, or services.	Parent / Child / Admin
Child profile creation with PIN and biometric option	Implemented in the Gamefide client and supported by backend routes, models, or services.	Parent / Child / Admin
Subject-wise game catalog	Implemented in the Gamefide client and supported by backend routes, models, or services.	Parent / Child / Admin
Technology 40-level quiz	Implemented in the Gamefide client and supported by backend routes, models, or services.	Parent / Child / Admin
AI Storybook with Gemini	Implemented in the Gamefide client and supported by backend routes, models, or services.	Parent / Child / Admin
Progress, points, streaks, achievements	Implemented in the Gamefide client and supported by backend routes, models, or services.	Parent / Child / Admin
Parent reporting with charts	Implemented in the Gamefide client and supported by backend routes, models, or services.	Parent / Child / Admin
Swagger API documentation	Implemented in the Gamefide client and supported by backend routes, models, or services.	Parent / Child / Admin

3.2 Functional Description

The user starts at the splash screen, selects a role, authenticates, and enters the relevant portal. Parents add and monitor children. Children login, select subjects, play games, submit scores, unlock levels, generate stories, and view achievements. The backend stores and returns all required data through REST APIs.

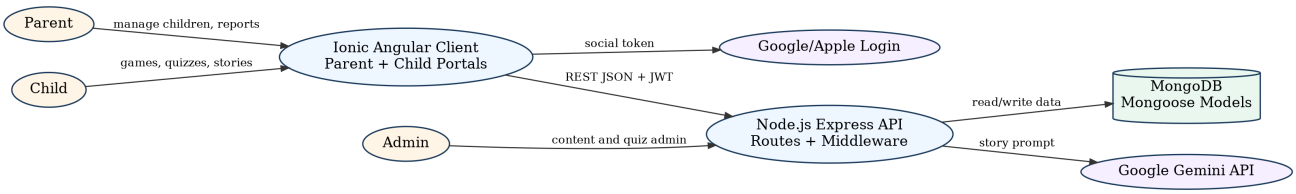


Figure 1: *High-level context diagram for Gamefide*

Table 7: User classes and characteristics

User Class	Characteristics	Responsibilities	Priority
Child	Aged 7 to 12; needs simple visual screens.	Play games, quizzes, stories, view badges.	High
Parent	Guardian managing child accounts.	Register, add children, view reports, settings.	High
Admin	Content and quiz manager.	Manage quiz/content data.	Medium
Developer	Maintainer.	Deploy, test, extend modules.	Medium

3.4 Constraints and Dependencies

- Ionic Angular frontend must remain responsive on mobile screens.
- Backend must use Node.js Express with MongoDB and Mongoose.
- JWT role checks are required for protected routes.
- Gemini API key is required for AI story generation.
- Quiz data must exist before level quiz can be attempted.
- Capacitor plugins depend on supported devices.

Chapter 4. System Requirements

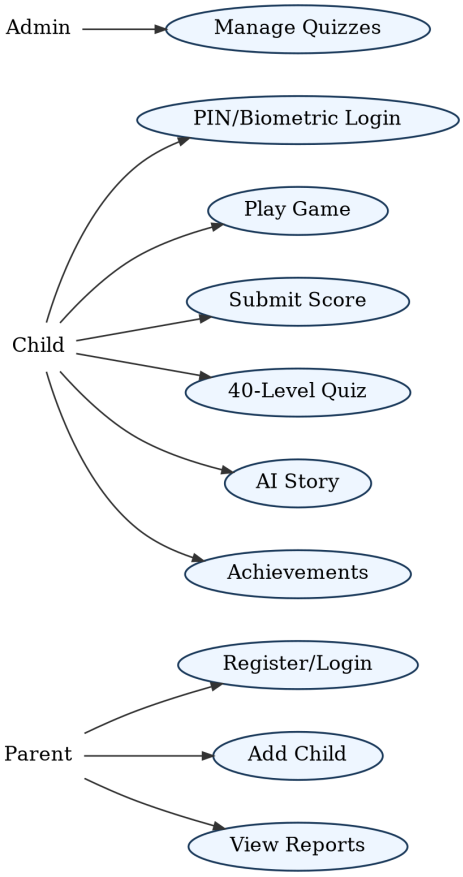


Figure 2: *Use case diagram for Gamefide*

4.1.1 Register Parent Account

Table 8: UC-1 overview - Register Parent Account

Identifier	UC-1
Purpose	To allow the actor to perform register parent account within Gamefide.
Priority	High
Pre-conditions	Valid user role, available route, and required data exist.
Post-conditions	The system stores, displays, or updates the expected result successfully.

Table 9: UC-1 typical course of action

S#	Actor Action	System Response
1	Actor starts register parent account workflow.	System displays the required page or API route.
2	Actor provides valid input.	System validates input and role.
3	Actor submits request.	System processes controller logic and database operation.
4	Actor receives feedback.	System returns success response, screen update, score, report, or message.

Table 10: UC-1 alternate course

S#	Alternate Action	System Response
1	Invalid or missing input is provided.	System returns validation message.
2	Unauthorized user attempts action.	System denies access and protects data.

4.1.2 Login Parent

Table 11: UC-2 overview - Login Parent

Identifier	UC-2

Purpose	To allow the actor to perform login parent within Gamefide.
Priority	High
Pre-conditions	Valid user role, available route, and required data exist.
Post-conditions	The system stores, displays, or updates the expected result successfully.

Table 12: UC-2 typical course of action

S#	Actor Action	System Response
1	Actor starts login parent workflow.	System displays the required page or API route.
2	Actor provides valid input.	System validates input and role.
3	Actor submits request.	System processes controller logic and database operation.
4	Actor receives feedback.	System returns success response, screen update, score, report, or message.

Table 13: UC-2 alternate course

S#	Alternate Action	System Response
1	Invalid or missing input is provided.	System returns validation message.
2	Unauthorized user attempts action.	System denies access and protects data.

4.1.3 Register Child Profile

Table 14: UC-3 overview - Register Child Profile

Identifier	UC-3
Purpose	To allow the actor to perform register child profile within Gamefide.
Priority	High
Pre-conditions	Valid user role, available route, and required data exist.
Post-conditions	The system stores, displays, or updates the expected result successfully.

Table 15: UC-3 typical course of action

S#	Actor Action	System Response
1	Actor starts register child profile workflow.	System displays the required page or API route.
2	Actor provides valid input.	System validates input and role.
3	Actor submits request.	System processes controller logic and database operation.
4	Actor receives feedback.	System returns success response, screen update, score, report, or message.

Table 16: UC-3 alternate course

S#	Alternate Action	System Response
1	Invalid or missing input is provided.	System returns validation message.
2	Unauthorized user attempts action.	System denies access and protects data.

4.1.4 Register Child with Fingerprint

Table 17: UC-4 overview - Register Child with Fingerprint

Identifier	UC-4
Purpose	To allow the actor to perform register child with fingerprint within Gamefide.
Priority	Medium
Pre-conditions	Valid user role, available route, and required data exist.
Post-conditions	The system stores, displays, or updates the expected result successfully.

Table 18: UC-4 typical course of action

S#	Actor Action	System Response
1	Actor starts register child with fingerprint workflow.	System displays the required page or API route.
2	Actor provides valid input.	System validates input and role.
3	Actor submits request.	System processes controller logic and database operation.

S#	Actor Action	System Response
4	Actor receives feedback.	System returns success response, screen update, score, report, or message.

Table 19: UC-4 alternate course

S#	Alternate Action	System Response
1	Invalid or missing input is provided.	System returns validation message.
2	Unauthorized user attempts action.	System denies access and protects data.

4.1.5 Child Login

Table 20: UC-5 overview - Child Login

Identifier	UC-5
Purpose	To allow the actor to perform child login within Gamefide.
Priority	High
Pre-conditions	Valid user role, available route, and required data exist.
Post-conditions	The system stores, displays, or updates the expected result successfully.

Table 21: UC-5 typical course of action

S#	Actor Action	System Response
1	Actor starts child login workflow.	System displays the required page or API route.
2	Actor provides valid input.	System validates input and role.
3	Actor submits request.	System processes controller logic and database operation.
4	Actor receives feedback.	System returns success response, screen update, score, report, or message.

Table 22: UC-5 alternate course

S#	Alternate Action	System Response
1	Invalid or missing input is provided.	System returns validation message.
2	Unauthorized user attempts action.	System denies access and protects data.

4.1.6 View Child Home Dashboard

Table 23: UC-6 overview - View Child Home Dashboard

Identifier	UC-6
Purpose	To allow the actor to perform view child home dashboard within Gamefide.
Priority	High
Pre-conditions	Valid user role, available route, and required data exist.
Post-conditions	The system stores, displays, or updates the expected result successfully.

Table 24: UC-6 typical course of action

S#	Actor Action	System Response
1	Actor starts view child home dashboard workflow.	System displays the required page or API route.
2	Actor provides valid input.	System validates input and role.
3	Actor submits request.	System processes controller logic and database operation.
4	Actor receives feedback.	System returns success response, screen update, score, report, or message.

Table 25: UC-6 alternate course

S#	Alternate Action	System Response
1	Invalid or missing input is provided.	System returns validation message.
2	Unauthorized user attempts action.	System denies access and protects data.

4.1.7 Select Subject

Table 26: UC-7 overview - Select Subject

Identifier	UC-7
Purpose	To allow the actor to perform select subject within Gamefide.
Priority	High
Pre-conditions	Valid user role, available route, and required data exist.
Post-conditions	The system stores, displays, or updates the expected result successfully.

Table 27: UC-7 typical course of action

S#	Actor Action	System Response
1	Actor starts select subject workflow.	System displays the required page or API route.
2	Actor provides valid input.	System validates input and role.
3	Actor submits request.	System processes controller logic and database operation.
4	Actor receives feedback.	System returns success response, screen update, score, report, or message.

Table 28: UC-7 alternate course

S#	Alternate Action	System Response
1	Invalid or missing input is provided.	System returns validation message.
2	Unauthorized user attempts action.	System denies access and protects data.

4.1.8 Play Educational Game

Table 29: UC-8 overview - Play Educational Game

Identifier	UC-8
Purpose	To allow the actor to perform play educational game within Gamefide.
Priority	High
Pre-conditions	Valid user role, available route, and required data exist.
Post-conditions	The system stores, displays, or updates the expected result successfully.

Table 30: UC-8 typical course of action

S#	Actor Action	System Response
1	Actor starts play educational game workflow.	System displays the required page or API route.
2	Actor provides valid input.	System validates input and role.
3	Actor submits request.	System processes controller logic and database operation.
4	Actor receives feedback.	System returns success response, screen update, score, report, or message.

Table 31: UC-8 alternate course

S#	Alternate Action	System Response
1	Invalid or missing input is provided.	System returns validation message.
2	Unauthorized user attempts action.	System denies access and protects data.

4.1.9 Submit Game Score

Table 32: UC-9 overview - Submit Game Score

Identifier	UC-9
Purpose	To allow the actor to perform submit game score within Gamefide.
Priority	High
Pre-conditions	Valid user role, available route, and required data exist.
Post-conditions	The system stores, displays, or updates the expected result successfully.

Table 33: UC-9 typical course of action

S#	Actor Action	System Response
1	Actor starts submit game score workflow.	System displays the required page or API route.
2	Actor provides valid input.	System validates input and role.

S#	Actor Action	System Response
3	Actor submits request.	System processes controller logic and database operation.
4	Actor receives feedback.	System returns success response, screen update, score, report, or message.

Table 34: UC-9 alternate course

S#	Alternate Action	System Response
1	Invalid or missing input is provided.	System returns validation message.
2	Unauthorized user attempts action.	System denies access and protects data.

4.1.10 Attempt Level Quiz

Table 35: UC-10 overview - Attempt Level Quiz

Identifier	UC-10
Purpose	To allow the actor to perform attempt level quiz within Gamefide.
Priority	High
Pre-conditions	Valid user role, available route, and required data exist.
Post-conditions	The system stores, displays, or updates the expected result successfully.

Table 36: UC-10 typical course of action

S#	Actor Action	System Response
1	Actor starts attempt level quiz workflow.	System displays the required page or API route.
2	Actor provides valid input.	System validates input and role.
3	Actor submits request.	System processes controller logic and database operation.
4	Actor receives feedback.	System returns success response, screen update, score, report, or message.

Table 37: UC-10 alternate course

S#	Alternate Action	System Response
1	Invalid or missing input is provided.	System returns validation message.
2	Unauthorized user attempts action.	System denies access and protects data.

4.1.11 Generate AI Story

Table 38: UC-11 overview - Generate AI Story

Identifier	UC-11
Purpose	To allow the actor to perform generate ai story within Gamefide.
Priority	High
Pre-conditions	Valid user role, available route, and required data exist.
Post-conditions	The system stores, displays, or updates the expected result successfully.

Table 39: UC-11 typical course of action

S#	Actor Action	System Response
1	Actor starts generate ai story workflow.	System displays the required page or API route.
2	Actor provides valid input.	System validates input and role.
3	Actor submits request.	System processes controller logic and database operation.
4	Actor receives feedback.	System returns success response, screen update, score, report, or message.

Table 40: UC-11 alternate course

S#	Alternate Action	System Response
1	Invalid or missing input is provided.	System returns validation message.
2	Unauthorized user attempts action.	System denies access and protects data.

4.1.12 View Achievements

Table 41: UC-12 overview - View Achievements

Identifier	UC-12
Purpose	To allow the actor to perform view achievements within Gamefide.
Priority	Medium
Pre-conditions	Valid user role, available route, and required data exist.
Post-conditions	The system stores, displays, or updates the expected result successfully.

Table 42: UC-12 typical course of action

S#	Actor Action	System Response
1	Actor starts view achievements workflow.	System displays the required page or API route.
2	Actor provides valid input.	System validates input and role.
3	Actor submits request.	System processes controller logic and database operation.
4	Actor receives feedback.	System returns success response, screen update, score, report, or message.

Table 43: UC-12 alternate course

S#	Alternate Action	System Response
1	Invalid or missing input is provided.	System returns validation message.
2	Unauthorized user attempts action.	System denies access and protects data.

4.1.13 View Parent Dashboard

Table 44: UC-13 overview - View Parent Dashboard

Identifier	UC-13
Purpose	To allow the actor to perform view parent dashboard within Gamefide.
Priority	High
Pre-conditions	Valid user role, available route, and required data exist.
Post-conditions	The system stores, displays, or updates the expected result successfully.

Table 45: UC-13 typical course of action

S#	Actor Action	System Response
1	Actor starts view parent dashboard workflow.	System displays the required page or API route.
2	Actor provides valid input.	System validates input and role.
3	Actor submits request.	System processes controller logic and database operation.
4	Actor receives feedback.	System returns success response, screen update, score, report, or message.

Table 46: UC-13 alternate course

S#	Alternate Action	System Response
1	Invalid or missing input is provided.	System returns validation message.
2	Unauthorized user attempts action.	System denies access and protects data.

4.1.14 View Reports

Table 47: UC-14 overview - View Reports

Identifier	UC-14
Purpose	To allow the actor to perform view reports within Gamefide.
Priority	High
Pre-conditions	Valid user role, available route, and required data exist.
Post-conditions	The system stores, displays, or updates the expected result successfully.

Table 48: UC-14 typical course of action

S#	Actor Action	System Response
1	Actor starts view reports workflow.	System displays the required page or API route.

S#	Actor Action	System Response
2	Actor provides valid input.	System validates input and role.
3	Actor submits request.	System processes controller logic and database operation.
4	Actor receives feedback.	System returns success response, screen update, score, report, or message.

Table 49: UC-14 alternate course

S#	Alternate Action	System Response
1	Invalid or missing input is provided.	System returns validation message.
2	Unauthorized user attempts action.	System denies access and protects data.

4.1.15 Manage Quiz Questions

Table 50: UC-15 overview - Manage Quiz Questions

Identifier	UC-15
Purpose	To allow the actor to perform manage quiz questions within Gamefide.
Priority	Medium
Pre-conditions	Valid user role, available route, and required data exist.
Post-conditions	The system stores, displays, or updates the expected result successfully.

Table 51: UC-15 typical course of action

S#	Actor Action	System Response
1	Actor starts manage quiz questions workflow.	System displays the required page or API route.
2	Actor provides valid input.	System validates input and role.
3	Actor submits request.	System processes controller logic and database operation.
4	Actor receives feedback.	System returns success response, screen update, score, report, or message.

Table 52: UC-15 alternate course

S#	Alternate Action	System Response
1	Invalid or missing input is provided.	System returns validation message.
2	Unauthorized user attempts action.	System denies access and protects data.

4.2 Requirements Traceability Matrix

Table 53: Requirements traceability matrix

Req ID	Use Case	Module	Verification	Priority
FR-1	Register Parent Account	Authentication	Functional, UI, and API test	High
FR-2	Login Parent	Authentication	Functional, UI, and API test	High
FR-3	Register Child Profile	Authentication	Functional, UI, and API test	High
FR-4	Register Child with Fingerprint	Authentication	Functional, UI, and API test	Medium
FR-5	Child Login	Authentication	Functional, UI, and API test	High
FR-6	View Child Home Dashboard	Parent/Child Portal	Functional, UI, and API test	High
FR-7	Select Subject	Parent/Child Portal	Functional, UI, and API test	High
FR-8	Play Educational Game	Game/Quiz	Functional, UI, and API test	High
FR-9	Submit Game Score	Game/Quiz	Functional, UI, and API test	High
FR-10	Attempt Level Quiz	Game/Quiz	Functional, UI, and API test	High
FR-11	Generate AI Story	AI Story	Functional, UI, and API test	High
FR-12	View Achievements	Parent/Child Portal	Functional, UI, and API test	Medium
FR-13	View Parent Dashboard	Parent/Child Portal	Functional, UI, and API test	High
FR-14	View Reports	Parent/Child Portal	Functional, UI, and API test	High
FR-15	Manage Quiz Questions	Parent/Child Portal	Functional, UI, and API test	Medium

4.3 Nonfunctional Requirements

Table 54: Nonfunctional requirements

Category	Requirement	Verification
Performance	Screens and APIs should respond quickly in local testing.	UI/API tests
Security	JWT and role checks must protect parent and child data.	Auth tests
Usability	Child screens must be simple, colorful, and readable.	Visual review
Reliability	Score updates must preserve valid progress.	Repeated score tests
Maintainability	Routes, controllers, models, and services should be separated.	Code review
Portability	App should run in browser and be packageable through Capacitor.	Ionic/Capacitor test

Chapter 5. Requirements Analysis and Modeling

Full ER diagram for MongoDB and Mongoose models

This diagram supports the analysis and modeling requirement by showing how the system structure, data, user actions, and components interact. It helps evaluators understand the implemented design before reading code details.

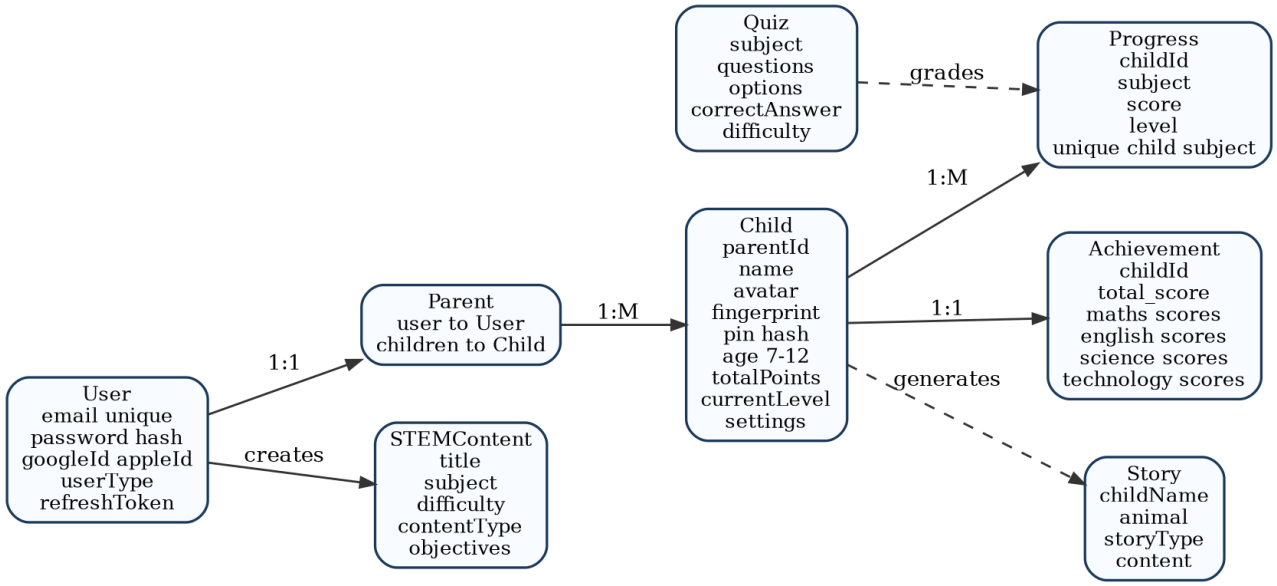


Figure 3: *Full ER diagram for MongoDB and Mongoose models*

UML class diagram for controllers and models

This diagram supports the analysis and modeling requirement by showing how the system structure, data, user actions, and components interact. It helps evaluators understand the implemented design before reading code details.

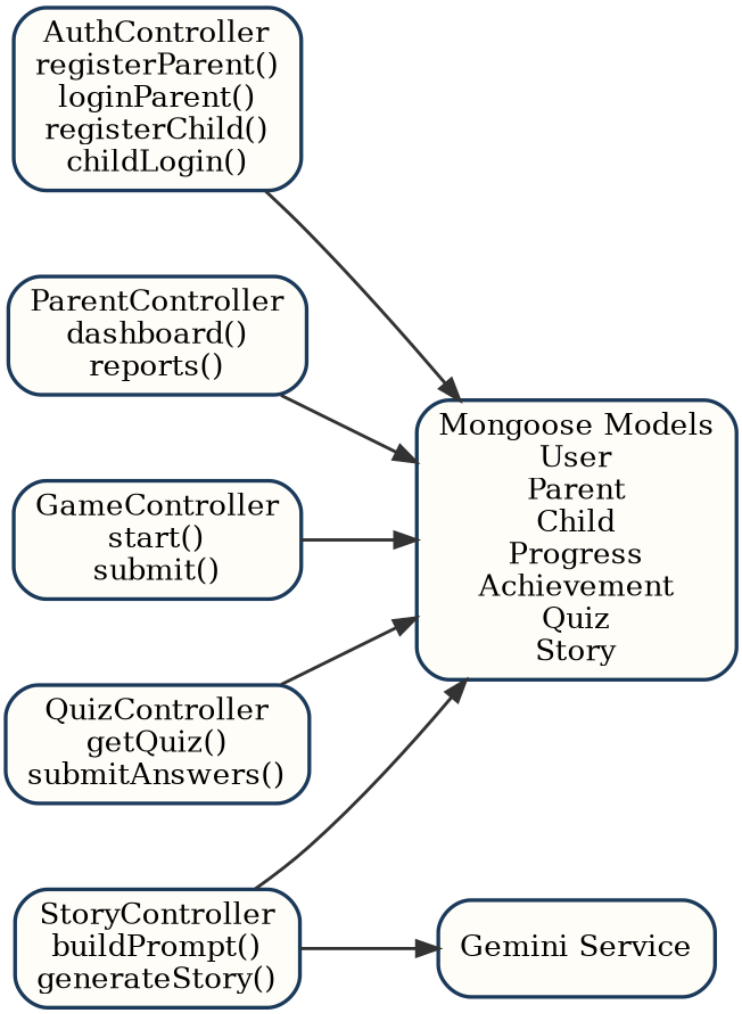


Figure 4: *UML class diagram for controllers and models*

Sequence diagram - parent registers child

This diagram supports the analysis and modeling requirement by showing how the system structure, data, user actions, and components interact. It helps evaluators understand the implemented design before reading code details.

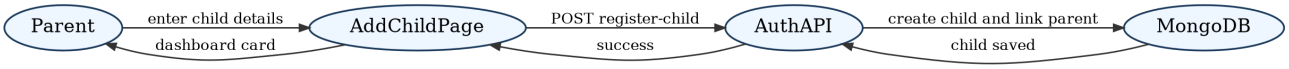


Figure 5: *Sequence diagram - parent registers child*

Sequence diagram - child plays game and submits score

This diagram supports the analysis and modeling requirement by showing how the system structure, data, user actions, and components interact. It helps evaluators understand the implemented design before reading code details.

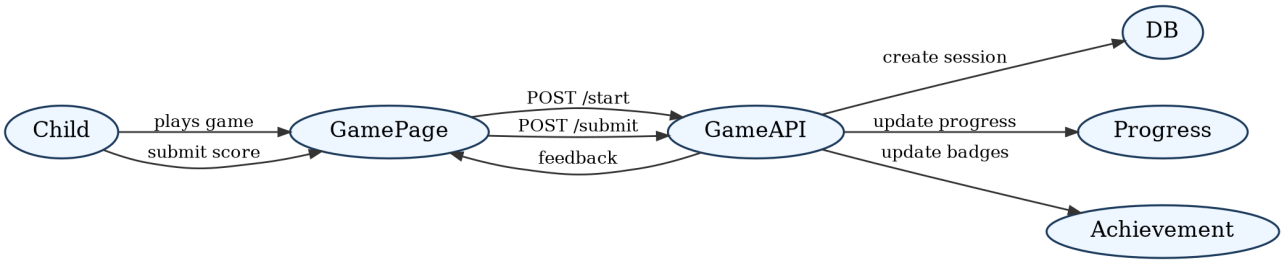


Figure 6: *Sequence diagram - child plays game and submits score*

Sequence diagram - AI story generation

This diagram supports the analysis and modeling requirement by showing how the system structure, data, user actions, and components interact. It helps evaluators understand the implemented design before reading code details.

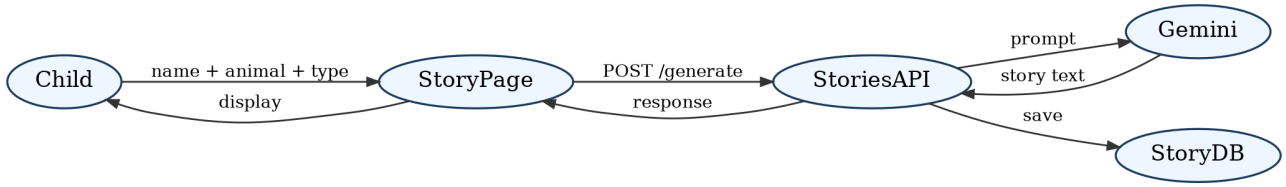


Figure 7: *Sequence diagram - AI story generation*

Sequence diagram - parent reports

This diagram supports the analysis and modeling requirement by showing how the system structure, data, user actions, and components interact. It helps evaluators understand the implemented design before reading code details.

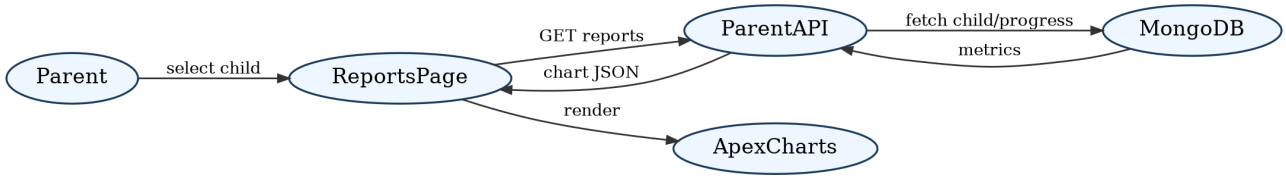


Figure 8: *Sequence diagram - parent reports*

DFD Level 0

This diagram supports the analysis and modeling requirement by showing how the system structure, data, user actions, and components interact. It helps evaluators understand the implemented design before reading code details.

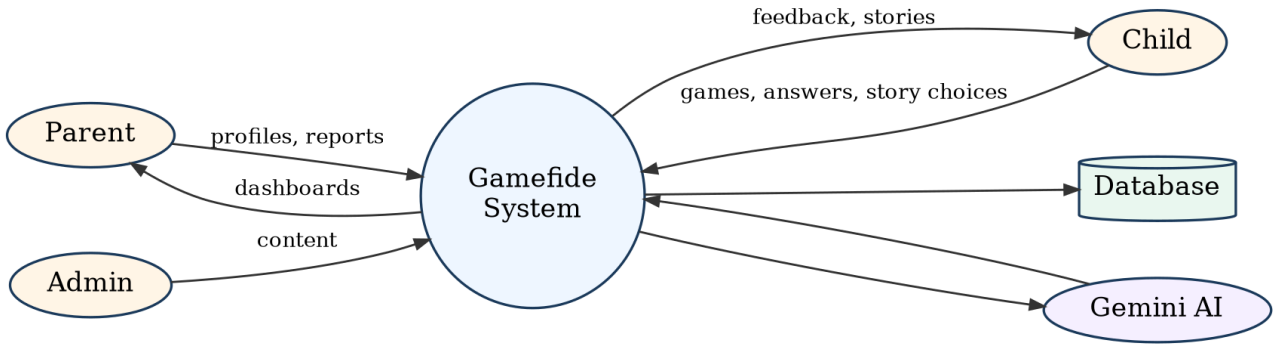


Figure 9: *DFD Level 0*

DFD Level 1

This diagram supports the analysis and modeling requirement by showing how the system structure, data, user actions, and components interact. It helps evaluators understand the implemented design before reading code details.

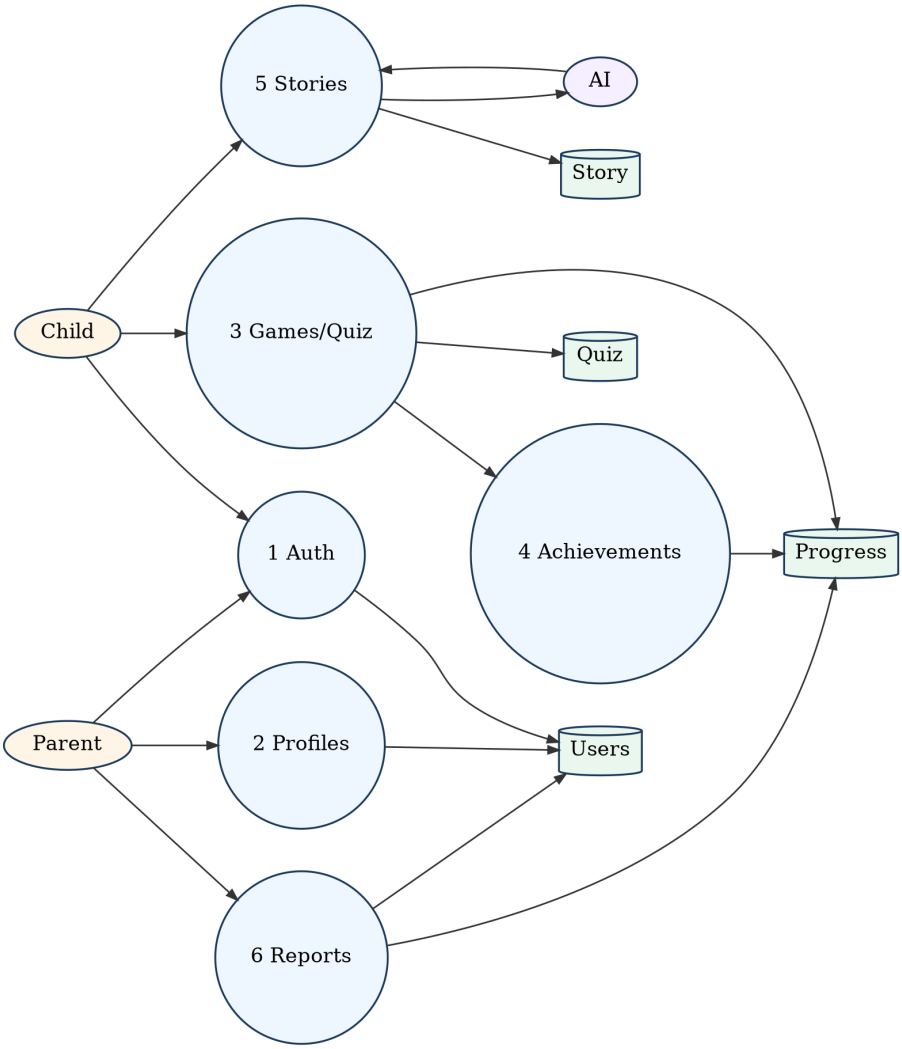


Figure 10: *DFD Level 1*

Child workflow activity diagram

This diagram supports the analysis and modeling requirement by showing how the system structure, data, user actions, and components interact. It helps evaluators understand the implemented design before reading code details.

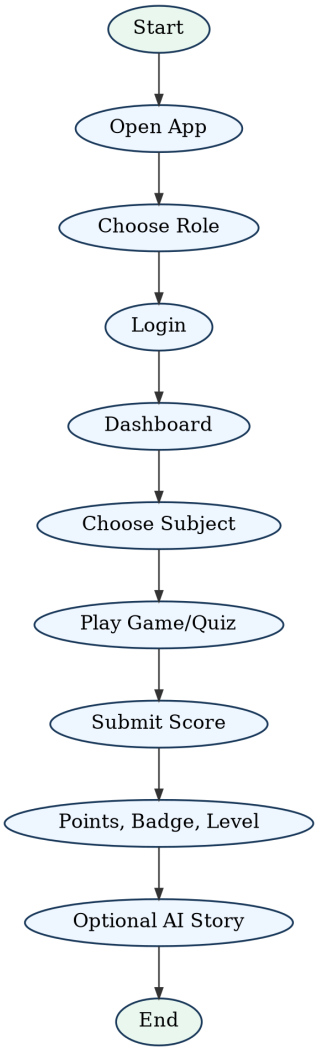


Figure 11: *Child workflow activity diagram*

Quiz level state diagram

This diagram supports the analysis and modeling requirement by showing how the system structure, data, user actions, and components interact. It helps evaluators understand the implemented design before reading code details.

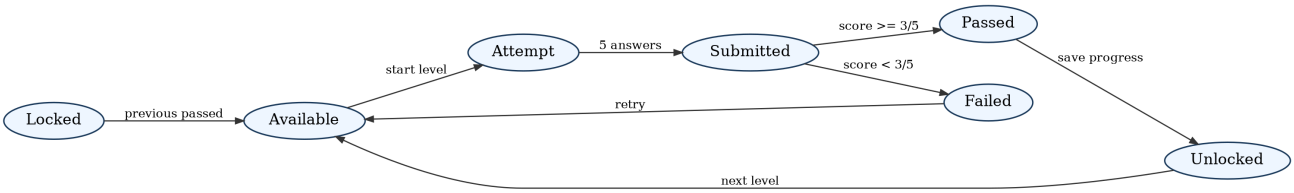


Figure 12: *Quiz level state diagram*

Table 55: Data dictionary

Model	Important Fields	Purpose
User	email, password hash, googleId, appleId, userType, refreshToken	User supports storage and retrieval of core Gamefide data.
Parent	user reference, children references	Parent supports storage and retrieval of core Gamefide data.
Child	parentId, name, avatar, fingerprint, pin hash, age, points, level, settings	Child supports storage and retrieval of core Gamefide data.
Progress	childId, subject, score, level, unique child-subject index	Progress supports storage and retrieval of core Gamefide data.
Achievement	childId, total_score, subject score arrays	Achievement supports storage and retrieval of core Gamefide data.
TotalAchievement	achievementId, total_achievement_score	TotalAchievement supports storage and retrieval of core Gamefide data.
Quiz	subject, questions, options, correctAnswer, difficulty	Quiz supports storage and retrieval of core Gamefide data.

Model	Important Fields	Purpose
STEMContent	title, description, subject, difficulty, contentType, objectives	STEMContent supports storage and retrieval of core Gamefide data.
Story	childName, animal, storyType, generated content	Story supports storage and retrieval of core Gamefide data.

Chapter 6. Technical Architecture

Component diagram

The diagram explains how frontend, backend, database, and external services collaborate in the selected architecture.

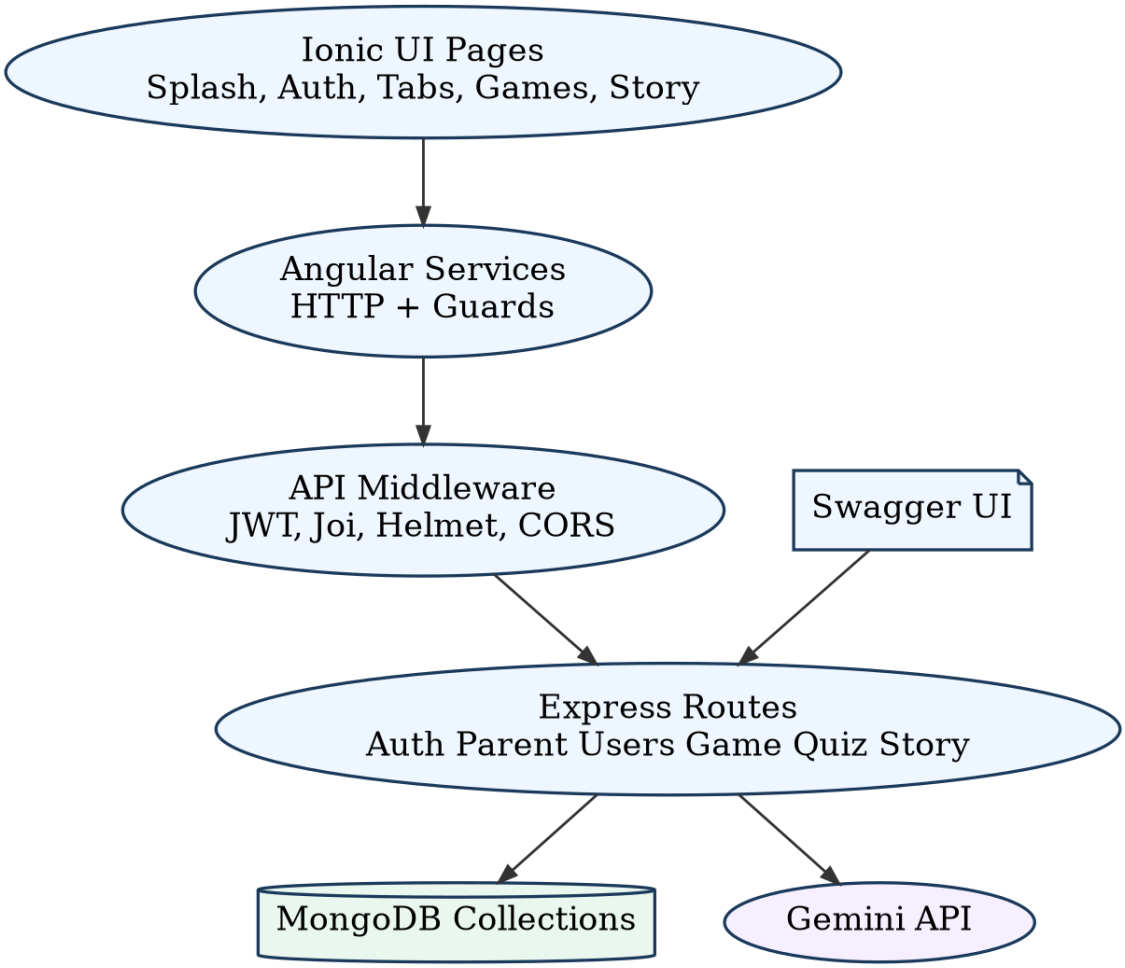


Figure 13: *Component diagram*

Deployment architecture

The diagram explains how frontend, backend, database, and external services collaborate in the selected architecture.

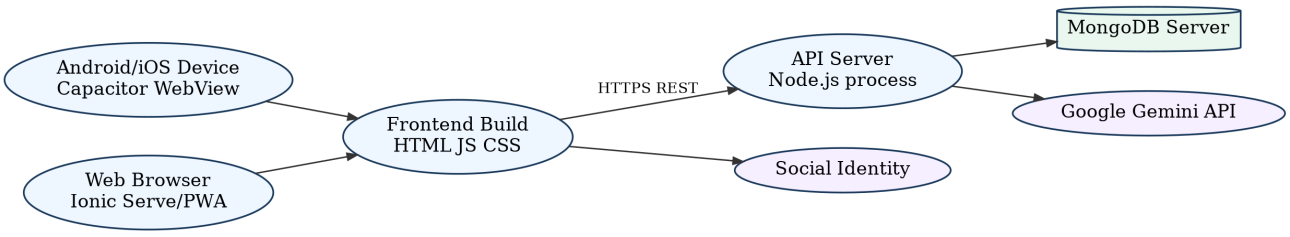


Figure 14: *Deployment architecture*

Table 56: Component interaction table

Component	Collaborates With	Input	Output
Ionic pages	Angular services	Forms/taps/routes	UI updates
Angular services	Express API	JSON + token	Response objects
JWT middleware	Protected routes	Authorization token	Authenticated user
Controllers	Mongoose models	Validated payload	Database result
Story controller	Gemini API	Prompt	Generated story

Component	Collaborates With	Input	Output
Parent controller	Progress data	Child ID	Chart metrics

6.3 Design Patterns

Table 57: Design patterns and reuse

Pattern	Use	Benefit
Layered architecture	Frontend/API/database separation	Maintainability
MVC-style controller routing	Routes call controllers and models	Organization
Middleware chain	JWT, Joi, Helmet, CORS	Consistent security
Service abstraction	Angular services wrap HTTP calls	Reusability
Gamification pattern	Points, streaks, badges, unlocking	Motivation
Prompt template	Story type controls Gemini prompt	Safer AI output

Chapter 7. Detailed Design and Implementation

Table 58: API endpoint reference

Endpoint	Method	Purpose
/api/auth/register	GET/POST/PUT/DELETE as required	Supports authentication, profiles, reports, progress, games, quizzes, achievements, or AI stories.
/api/auth/login-parent	GET/POST/PUT/DELETE as required	Supports authentication, profiles, reports, progress, games, quizzes, achievements, or AI stories.
/api/auth/register-child	GET/POST/PUT/DELETE as required	Supports authentication, profiles, reports, progress, games, quizzes, achievements, or AI stories.
/api/auth/child-login	GET/POST/PUT/DELETE as required	Supports authentication, profiles, reports, progress, games, quizzes, achievements, or AI stories.
/api/parent/dashboard	GET/POST/PUT/DELETE as required	Supports authentication, profiles, reports, progress, games, quizzes, achievements, or AI stories.
/api/parent/reports/:childId	GET/POST/PUT/DELETE as required	Supports authentication, profiles, reports, progress, games, quizzes, achievements, or AI stories.
/api/users/children	GET/POST/PUT/DELETE as required	Supports authentication, profiles, reports, progress, games, quizzes, achievements, or AI stories.
/api/progress	GET/POST/PUT/DELETE as required	Supports authentication, profiles, reports, progress, games, quizzes, achievements, or AI stories.
/api/achievements	GET/POST/PUT/DELETE as required	Supports authentication, profiles, reports, progress, games, quizzes, achievements, or AI stories.
/api/quiz/:subject	GET/POST/PUT/DELETE as required	Supports authentication, profiles, reports, progress, games, quizzes, achievements, or AI stories.
/api/quiz/submit	GET/POST/PUT/DELETE as required	Supports authentication, profiles, reports, progress, games, quizzes, achievements, or AI stories.
/api/game/start	GET/POST/PUT/DELETE as required	Supports authentication, profiles, reports, progress, games, quizzes, achievements, or AI stories.
/api/game/submit	GET/POST/PUT/DELETE as required	Supports authentication, profiles, reports, progress, games, quizzes, achievements, or AI stories.
/api/stories/generate	GET/POST/PUT/DELETE as required	Supports authentication, profiles, reports, progress, games, quizzes, achievements, or AI stories.

7.2 Human Interface Design

Table 59: Human interface inventory

Screen	Input	Output	Design Note
Splash	Mode tap	Next route	Friendly first impression
Parent login	Email/social login	Dashboard	Secure and simple
Add child	Name, age, PIN, avatar	Child card	Parent-controlled setup
Child home	Navigation taps	Points and level	Motivational
Subjects	Card tap	Subject games	Visual navigation
Game screen	Tap/drag/answer	Score feedback	Learning by doing
Story page	Name/animal/type	Generated story	AI reading activity

Screen	Input	Output	Design Note
Reports	Child selection	Charts	Parent monitoring

7.3 AI Prompt Design

The backend does not allow the child client to send arbitrary prompts directly to Gemini. Instead, it creates controlled story prompts from child name, animal, and story type. Bedtime prompts require calm short stories, math prompts include a simple addition or subtraction challenge, and moral prompts end with a clear moral lesson.

```
POST /api/stories/generate
{ childName, animal, storyType }
Backend -> build controlled prompt -> Gemini API -> save Story -> return content
```

Chapter 8. Educational Games - Detailed Sub-Chapters

8.1 Mathematics Games

Mathematics games transform curriculum concepts into visual, interactive actions. Each game is mapped to score submission, achievements, and parent reporting so learning activity becomes measurable.

Mathematics-1: Balloon Pop

Balloon Pop is a dedicated mathematics game module. It gives children a clear task, immediate feedback, and a score outcome that can be stored through the common backend progress and achievement workflow. The design uses simple instructions, colorful cards, and age-appropriate interaction so learners can focus on the concept instead of complex controls.

Table 60: Balloon Pop design sheet

Design Item	Specification
Learning objective	Improve mathematics understanding through balloon pop activities.
User interaction	Tap, drag, select, match, order, or navigate depending on the game.
Scoring	Correct actions increase score and incorrect actions provide retry feedback.
Data recorded	childId, subject, game_name, score, level, achievement category.
Future improvement	Adaptive difficulty, narration, richer animations, teacher content, and more analytics.

Mathematics-2: Counting Fun

Counting Fun is a dedicated mathematics game module. It gives children a clear task, immediate feedback, and a score outcome that can be stored through the common backend progress and achievement workflow. The design uses simple instructions, colorful cards, and age-appropriate interaction so learners can focus on the concept instead of complex controls.

Table 61: Counting Fun design sheet

Design Item	Specification
Learning objective	Improve mathematics understanding through counting fun activities.
User interaction	Tap, drag, select, match, order, or navigate depending on the game.
Scoring	Correct actions increase score and incorrect actions provide retry feedback.
Data recorded	childId, subject, game_name, score, level, achievement category.
Future improvement	Adaptive difficulty, narration, richer animations, teacher content, and more analytics.

Mathematics-3: Pattern Match

Pattern Match is a dedicated mathematics game module. It gives children a clear task, immediate feedback, and a score outcome that can be stored through the common backend progress and achievement workflow. The design uses simple instructions, colorful cards, and age-appropriate interaction so learners can focus on the concept instead of complex controls.

Table 62: Pattern Match design sheet

Design Item	Specification
Learning objective	Improve mathematics understanding through pattern match activities.
User interaction	Tap, drag, select, match, order, or navigate depending on the game.
Scoring	Correct actions increase score and incorrect actions provide retry feedback.
Data recorded	childId, subject, game_name, score, level, achievement category.
Future improvement	Adaptive difficulty, narration, richer animations, teacher content, and more analytics.

Mathematics-4: Clock Reading

Clock Reading is a dedicated mathematics game module. It gives children a clear task, immediate feedback, and a score outcome that can be stored through the common backend progress and achievement workflow. The design uses simple instructions, colorful cards, and age-appropriate interaction so learners can focus on the concept instead of complex controls.

Table 63: Clock Reading design sheet

Design Item	Specification
Learning objective	Improve mathematics understanding through clock reading activities.
User interaction	Tap, drag, select, match, order, or navigate depending on the game.
Scoring	Correct actions increase score and incorrect actions provide retry feedback.

Design Item	Specification
Data recorded	childId, subject, game_name, score, level, achievement category.
Future improvement	Adaptive difficulty, narration, richer animations, teacher content, and more analytics.

Mathematics-5: Math Maze

Math Maze is a dedicated mathematics game module. It gives children a clear task, immediate feedback, and a score outcome that can be stored through the common backend progress and achievement workflow. The design uses simple instructions, colorful cards, and age-appropriate interaction so learners can focus on the concept instead of complex controls.

Table 64: Math Maze design sheet

Design Item	Specification
Learning objective	Improve mathematics understanding through math maze activities.
User interaction	Tap, drag, select, match, order, or navigate depending on the game.
Scoring	Correct actions increase score and incorrect actions provide retry feedback.
Data recorded	childId, subject, game_name, score, level, achievement category.
Future improvement	Adaptive difficulty, narration, richer animations, teacher content, and more analytics.

Mathematics-6: Fraction Match

Fraction Match is a dedicated mathematics game module. It gives children a clear task, immediate feedback, and a score outcome that can be stored through the common backend progress and achievement workflow. The design uses simple instructions, colorful cards, and age-appropriate interaction so learners can focus on the concept instead of complex controls.

Table 65: Fraction Match design sheet

Design Item	Specification
Learning objective	Improve mathematics understanding through fraction match activities.
User interaction	Tap, drag, select, match, order, or navigate depending on the game.
Scoring	Correct actions increase score and incorrect actions provide retry feedback.
Data recorded	childId, subject, game_name, score, level, achievement category.
Future improvement	Adaptive difficulty, narration, richer animations, teacher content, and more analytics.

Mathematics-7: Multiplication Quiz

Multiplication Quiz is a dedicated mathematics game module. It gives children a clear task, immediate feedback, and a score outcome that can be stored through the common backend progress and achievement workflow. The design uses simple instructions, colorful cards, and age-appropriate interaction so learners can focus on the concept instead of complex controls.

Table 66: Multiplication Quiz design sheet

Design Item	Specification
Learning objective	Improve mathematics understanding through multiplication quiz activities.
User interaction	Tap, drag, select, match, order, or navigate depending on the game.
Scoring	Correct actions increase score and incorrect actions provide retry feedback.
Data recorded	childId, subject, game_name, score, level, achievement category.
Future improvement	Adaptive difficulty, narration, richer animations, teacher content, and more analytics.

8.2 Science Games

Science games transform curriculum concepts into visual, interactive actions. Each game is mapped to score submission, achievements, and parent reporting so learning activity becomes measurable.

Science-1: Human Body Match

Human Body Match is a dedicated science game module. It gives children a clear task, immediate feedback, and a score outcome that can be stored through the common backend progress and achievement workflow. The design uses simple instructions, colorful cards, and age-appropriate interaction so learners can focus on the concept instead of complex controls.

Table 67: Human Body Match design sheet

Design Item	Specification
Learning objective	Improve science understanding through human body match activities.

Design Item	Specification
User interaction	Tap, drag, select, match, order, or navigate depending on the game.
Scoring	Correct actions increase score and incorrect actions provide retry feedback.
Data recorded	childId, subject, game_name, score, level, achievement category.
Future improvement	Adaptive difficulty, narration, richer animations, teacher content, and more analytics.

Science-2: Life Cycle Puzzle

Life Cycle Puzzle is a dedicated science game module. It gives children a clear task, immediate feedback, and a score outcome that can be stored through the common backend progress and achievement workflow. The design uses simple instructions, colorful cards, and age-appropriate interaction so learners can focus on the concept instead of complex controls.

Table 68: Life Cycle Puzzle design sheet

Design Item	Specification
Learning objective	Improve science understanding through life cycle puzzle activities.
User interaction	Tap, drag, select, match, order, or navigate depending on the game.
Scoring	Correct actions increase score and incorrect actions provide retry feedback.
Data recorded	childId, subject, game_name, score, level, achievement category.
Future improvement	Adaptive difficulty, narration, richer animations, teacher content, and more analytics.

Science-3: Weather Guessing

Weather Guessing is a dedicated science game module. It gives children a clear task, immediate feedback, and a score outcome that can be stored through the common backend progress and achievement workflow. The design uses simple instructions, colorful cards, and age-appropriate interaction so learners can focus on the concept instead of complex controls.

Table 69: Weather Guessing design sheet

Design Item	Specification
Learning objective	Improve science understanding through weather guessing activities.
User interaction	Tap, drag, select, match, order, or navigate depending on the game.
Scoring	Correct actions increase score and incorrect actions provide retry feedback.
Data recorded	childId, subject, game_name, score, level, achievement category.
Future improvement	Adaptive difficulty, narration, richer animations, teacher content, and more analytics.

Science-4: Animal Sounds and Habitats

Animal Sounds and Habitats is a dedicated science game module. It gives children a clear task, immediate feedback, and a score outcome that can be stored through the common backend progress and achievement workflow. The design uses simple instructions, colorful cards, and age-appropriate interaction so learners can focus on the concept instead of complex controls.

Table 70: Animal Sounds and Habitats design sheet

Design Item	Specification
Learning objective	Improve science understanding through animal sounds and habitats activities.
User interaction	Tap, drag, select, match, order, or navigate depending on the game.
Scoring	Correct actions increase score and incorrect actions provide retry feedback.
Data recorded	childId, subject, game_name, score, level, achievement category.
Future improvement	Adaptive difficulty, narration, richer animations, teacher content, and more analytics.

Science-5: Plant Parts

Plant Parts is a dedicated science game module. It gives children a clear task, immediate feedback, and a score outcome that can be stored through the common backend progress and achievement workflow. The design uses simple instructions, colorful cards, and age-appropriate interaction so learners can focus on the concept instead of complex controls.

Table 71: Plant Parts design sheet

Design Item	Specification
Learning objective	Improve science understanding through plant parts activities.
User interaction	Tap, drag, select, match, order, or navigate depending on the game.
Scoring	Correct actions increase score and incorrect actions provide retry feedback.

Design Item	Specification
Data recorded	childId, subject, game_name, score, level, achievement category.
Future improvement	Adaptive difficulty, narration, richer animations, teacher content, and more analytics.

Science-6: Sink or Float

Sink or Float is a dedicated science game module. It gives children a clear task, immediate feedback, and a score outcome that can be stored through the common backend progress and achievement workflow. The design uses simple instructions, colorful cards, and age-appropriate interaction so learners can focus on the concept instead of complex controls.

Table 72: Sink or Float design sheet

Design Item	Specification
Learning objective	Improve science understanding through sink or float activities.
User interaction	Tap, drag, select, match, order, or navigate depending on the game.
Scoring	Correct actions increase score and incorrect actions provide retry feedback.
Data recorded	childId, subject, game_name, score, level, achievement category.
Future improvement	Adaptive difficulty, narration, richer animations, teacher content, and more analytics.

Science-7: Water Cycle

Water Cycle is a dedicated science game module. It gives children a clear task, immediate feedback, and a score outcome that can be stored through the common backend progress and achievement workflow. The design uses simple instructions, colorful cards, and age-appropriate interaction so learners can focus on the concept instead of complex controls.

Table 73: Water Cycle design sheet

Design Item	Specification
Learning objective	Improve science understanding through water cycle activities.
User interaction	Tap, drag, select, match, order, or navigate depending on the game.
Scoring	Correct actions increase score and incorrect actions provide retry feedback.
Data recorded	childId, subject, game_name, score, level, achievement category.
Future improvement	Adaptive difficulty, narration, richer animations, teacher content, and more analytics.

Science-8: Science Tools

Science Tools is a dedicated science game module. It gives children a clear task, immediate feedback, and a score outcome that can be stored through the common backend progress and achievement workflow. The design uses simple instructions, colorful cards, and age-appropriate interaction so learners can focus on the concept instead of complex controls.

Table 74: Science Tools design sheet

Design Item	Specification
Learning objective	Improve science understanding through science tools activities.
User interaction	Tap, drag, select, match, order, or navigate depending on the game.
Scoring	Correct actions increase score and incorrect actions provide retry feedback.
Data recorded	childId, subject, game_name, score, level, achievement category.
Future improvement	Adaptive difficulty, narration, richer animations, teacher content, and more analytics.

Science-9: States of Matter

States of Matter is a dedicated science game module. It gives children a clear task, immediate feedback, and a score outcome that can be stored through the common backend progress and achievement workflow. The design uses simple instructions, colorful cards, and age-appropriate interaction so learners can focus on the concept instead of complex controls.

Table 75: States of Matter design sheet

Design Item	Specification
Learning objective	Improve science understanding through states of matter activities.
User interaction	Tap, drag, select, match, order, or navigate depending on the game.
Scoring	Correct actions increase score and incorrect actions provide retry feedback.
Data recorded	childId, subject, game_name, score, level, achievement category.
Future improvement	Adaptive difficulty, narration, richer animations, teacher content, and more analytics.

8.3 Technology Games

Technology games transform curriculum concepts into visual, interactive actions. Each game is mapped to score submission, achievements, and parent reporting so learning activity becomes measurable.

Technology-1: Computer Parts Match

Computer Parts Match is a dedicated technology game module. It gives children a clear task, immediate feedback, and a score outcome that can be stored through the common backend progress and achievement workflow. The design uses simple instructions, colorful cards, and age-appropriate interaction so learners can focus on the concept instead of complex controls.

Table 76: Computer Parts Match design sheet

Design Item	Specification
Learning objective	Improve technology understanding through computer parts match activities.
User interaction	Tap, drag, select, match, order, or navigate depending on the game.
Scoring	Correct actions increase score and incorrect actions provide retry feedback.
Data recorded	childId, subject, game_name, score, level, achievement category.
Future improvement	Adaptive difficulty, narration, richer animations, teacher content, and more analytics.

Technology-2: Robot Path Puzzle

Robot Path Puzzle is a dedicated technology game module. It gives children a clear task, immediate feedback, and a score outcome that can be stored through the common backend progress and achievement workflow. The design uses simple instructions, colorful cards, and age-appropriate interaction so learners can focus on the concept instead of complex controls.

Table 77: Robot Path Puzzle design sheet

Design Item	Specification
Learning objective	Improve technology understanding through robot path puzzle activities.
User interaction	Tap, drag, select, match, order, or navigate depending on the game.
Scoring	Correct actions increase score and incorrect actions provide retry feedback.
Data recorded	childId, subject, game_name, score, level, achievement category.
Future improvement	Adaptive difficulty, narration, richer animations, teacher content, and more analytics.

Technology-3: Keyboard Key Match

Keyboard Key Match is a dedicated technology game module. It gives children a clear task, immediate feedback, and a score outcome that can be stored through the common backend progress and achievement workflow. The design uses simple instructions, colorful cards, and age-appropriate interaction so learners can focus on the concept instead of complex controls.

Table 78: Keyboard Key Match design sheet

Design Item	Specification
Learning objective	Improve technology understanding through keyboard key match activities.
User interaction	Tap, drag, select, match, order, or navigate depending on the game.
Scoring	Correct actions increase score and incorrect actions provide retry feedback.
Data recorded	childId, subject, game_name, score, level, achievement category.
Future improvement	Adaptive difficulty, narration, richer animations, teacher content, and more analytics.

Technology-4: Internet Safety Quiz

Internet Safety Quiz is a dedicated technology game module. It gives children a clear task, immediate feedback, and a score outcome that can be stored through the common backend progress and achievement workflow. The design uses simple instructions, colorful cards, and age-appropriate interaction so learners can focus on the concept instead of complex controls.

Table 79: Internet Safety Quiz design sheet

Design Item	Specification
Learning objective	Improve technology understanding through internet safety quiz activities.
User interaction	Tap, drag, select, match, order, or navigate depending on the game.
Scoring	Correct actions increase score and incorrect actions provide retry feedback.
Data recorded	childId, subject, game_name, score, level, achievement category.

Design Item	Specification
Future improvement	Adaptive difficulty, narration, richer animations, teacher content, and more analytics.

Technology-5: Binary Number Puzzle

Binary Number Puzzle is a dedicated technology game module. It gives children a clear task, immediate feedback, and a score outcome that can be stored through the common backend progress and achievement workflow. The design uses simple instructions, colorful cards, and age-appropriate interaction so learners can focus on the concept instead of complex controls.

Table 80: Binary Number Puzzle design sheet

Design Item	Specification
Learning objective	Improve technology understanding through binary number puzzle activities.
User interaction	Tap, drag, select, match, order, or navigate depending on the game.
Scoring	Correct actions increase score and incorrect actions provide retry feedback.
Data recorded	childId, subject, game_name, score, level, achievement category.
Future improvement	Adaptive difficulty, narration, richer animations, teacher content, and more analytics.

Technology-6: AI vs Human Sorting

AI vs Human Sorting is a dedicated technology game module. It gives children a clear task, immediate feedback, and a score outcome that can be stored through the common backend progress and achievement workflow. The design uses simple instructions, colorful cards, and age-appropriate interaction so learners can focus on the concept instead of complex controls.

Table 81: AI vs Human Sorting design sheet

Design Item	Specification
Learning objective	Improve technology understanding through ai vs human sorting activities.
User interaction	Tap, drag, select, match, order, or navigate depending on the game.
Scoring	Correct actions increase score and incorrect actions provide retry feedback.
Data recorded	childId, subject, game_name, score, level, achievement category.
Future improvement	Adaptive difficulty, narration, richer animations, teacher content, and more analytics.

Technology-7: Tech Timeline Puzzle

Tech Timeline Puzzle is a dedicated technology game module. It gives children a clear task, immediate feedback, and a score outcome that can be stored through the common backend progress and achievement workflow. The design uses simple instructions, colorful cards, and age-appropriate interaction so learners can focus on the concept instead of complex controls.

Table 82: Tech Timeline Puzzle design sheet

Design Item	Specification
Learning objective	Improve technology understanding through tech timeline puzzle activities.
User interaction	Tap, drag, select, match, order, or navigate depending on the game.
Scoring	Correct actions increase score and incorrect actions provide retry feedback.
Data recorded	childId, subject, game_name, score, level, achievement category.
Future improvement	Adaptive difficulty, narration, richer animations, teacher content, and more analytics.

Technology-8: Gadget Recognition Game

Gadget Recognition Game is a dedicated technology game module. It gives children a clear task, immediate feedback, and a score outcome that can be stored through the common backend progress and achievement workflow. The design uses simple instructions, colorful cards, and age-appropriate interaction so learners can focus on the concept instead of complex controls.

Table 83: Gadget Recognition Game design sheet

Design Item	Specification
Learning objective	Improve technology understanding through gadget recognition game activities.
User interaction	Tap, drag, select, match, order, or navigate depending on the game.
Scoring	Correct actions increase score and incorrect actions provide retry feedback.
Data recorded	childId, subject, game_name, score, level, achievement category.
Future improvement	Adaptive difficulty, narration, richer animations, teacher content, and more analytics.

8.4 English Games

English games transform curriculum concepts into visual, interactive actions. Each game is mapped to score submission, achievements, and parent reporting so learning activity becomes measurable.

English-1: Alphabet Matching

Alphabet Matching is a dedicated english game module. It gives children a clear task, immediate feedback, and a score outcome that can be stored through the common backend progress and achievement workflow. The design uses simple instructions, colorful cards, and age-appropriate interaction so learners can focus on the concept instead of complex controls.

Table 84: Alphabet Matching design sheet

Design Item	Specification
Learning objective	Improve english understanding through alphabet matching activities.
User interaction	Tap, drag, select, match, order, or navigate depending on the game.
Scoring	Correct actions increase score and incorrect actions provide retry feedback.
Data recorded	childId, subject, game_name, score, level, achievement category.
Future improvement	Adaptive difficulty, narration, richer animations, teacher content, and more analytics.

English-2: Spelling Puzzle

Spelling Puzzle is a dedicated english game module. It gives children a clear task, immediate feedback, and a score outcome that can be stored through the common backend progress and achievement workflow. The design uses simple instructions, colorful cards, and age-appropriate interaction so learners can focus on the concept instead of complex controls.

Table 85: Spelling Puzzle design sheet

Design Item	Specification
Learning objective	Improve english understanding through spelling puzzle activities.
User interaction	Tap, drag, select, match, order, or navigate depending on the game.
Scoring	Correct actions increase score and incorrect actions provide retry feedback.
Data recorded	childId, subject, game_name, score, level, achievement category.
Future improvement	Adaptive difficulty, narration, richer animations, teacher content, and more analytics.

English-3: Word and Picture Match

Word and Picture Match is a dedicated english game module. It gives children a clear task, immediate feedback, and a score outcome that can be stored through the common backend progress and achievement workflow. The design uses simple instructions, colorful cards, and age-appropriate interaction so learners can focus on the concept instead of complex controls.

Table 86: Word and Picture Match design sheet

Design Item	Specification
Learning objective	Improve english understanding through word and picture match activities.
User interaction	Tap, drag, select, match, order, or navigate depending on the game.
Scoring	Correct actions increase score and incorrect actions provide retry feedback.
Data recorded	childId, subject, game_name, score, level, achievement category.
Future improvement	Adaptive difficulty, narration, richer animations, teacher content, and more analytics.

English-4: Fill in the Blanks

Fill in the Blanks is a dedicated english game module. It gives children a clear task, immediate feedback, and a score outcome that can be stored through the common backend progress and achievement workflow. The design uses simple instructions, colorful cards, and age-appropriate interaction so learners can focus on the concept instead of complex controls.

Table 87: Fill in the Blanks design sheet

Design Item	Specification
Learning objective	Improve english understanding through fill in the blanks activities.
User interaction	Tap, drag, select, match, order, or navigate depending on the game.
Scoring	Correct actions increase score and incorrect actions provide retry feedback.
Data recorded	childId, subject, game_name, score, level, achievement category.
Future improvement	Adaptive difficulty, narration, richer animations, teacher content, and more analytics.

English-5: Sentence Builder

Sentence Builder is a dedicated english game module. It gives children a clear task, immediate feedback, and a score outcome that can be stored through the common backend progress and achievement workflow. The design uses simple instructions, colorful cards, and age-appropriate interaction so learners can focus on the concept instead of complex controls.

Table 88: Sentence Builder design sheet

Design Item	Specification
Learning objective	Improve english understanding through sentence builder activities.
User interaction	Tap, drag, select, match, order, or navigate depending on the game.
Scoring	Correct actions increase score and incorrect actions provide retry feedback.
Data recorded	childId, subject, game_name, score, level, achievement category.
Future improvement	Adaptive difficulty, narration, richer animations, teacher content, and more analytics.

English-6: Rhyming Words

Rhyming Words is a dedicated english game module. It gives children a clear task, immediate feedback, and a score outcome that can be stored through the common backend progress and achievement workflow. The design uses simple instructions, colorful cards, and age-appropriate interaction so learners can focus on the concept instead of complex controls.

Table 89: Rhyming Words design sheet

Design Item	Specification
Learning objective	Improve english understanding through rhyming words activities.
User interaction	Tap, drag, select, match, order, or navigate depending on the game.
Scoring	Correct actions increase score and incorrect actions provide retry feedback.
Data recorded	childId, subject, game_name, score, level, achievement category.
Future improvement	Adaptive difficulty, narration, richer animations, teacher content, and more analytics.

English-7: Grammar Quiz

Grammar Quiz is a dedicated english game module. It gives children a clear task, immediate feedback, and a score outcome that can be stored through the common backend progress and achievement workflow. The design uses simple instructions, colorful cards, and age-appropriate interaction so learners can focus on the concept instead of complex controls.

Table 90: Grammar Quiz design sheet

Design Item	Specification
Learning objective	Improve english understanding through grammar quiz activities.
User interaction	Tap, drag, select, match, order, or navigate depending on the game.
Scoring	Correct actions increase score and incorrect actions provide retry feedback.
Data recorded	childId, subject, game_name, score, level, achievement category.
Future improvement	Adaptive difficulty, narration, richer animations, teacher content, and more analytics.

English-8: Story Ordering

Story Ordering is a dedicated english game module. It gives children a clear task, immediate feedback, and a score outcome that can be stored through the common backend progress and achievement workflow. The design uses simple instructions, colorful cards, and age-appropriate interaction so learners can focus on the concept instead of complex controls.

Table 91: Story Ordering design sheet

Design Item	Specification
Learning objective	Improve english understanding through story ordering activities.
User interaction	Tap, drag, select, match, order, or navigate depending on the game.
Scoring	Correct actions increase score and incorrect actions provide retry feedback.
Data recorded	childId, subject, game_name, score, level, achievement category.
Future improvement	Adaptive difficulty, narration, richer animations, teacher content, and more analytics.

English-9: Sight Word Memory

Sight Word Memory is a dedicated english game module. It gives children a clear task, immediate feedback, and a score outcome that can be stored through the common backend progress and achievement workflow. The design uses simple

instructions, colorful cards, and age-appropriate interaction so learners can focus on the concept instead of complex controls.

Table 92: Sight Word Memory design sheet

Design Item	Specification
Learning objective	Improve english understanding through sight word memory activities.
User interaction	Tap, drag, select, match, order, or navigate depending on the game.
Scoring	Correct actions increase score and incorrect actions provide retry feedback.
Data recorded	childId, subject, game_name, score, level, achievement category.
Future improvement	Adaptive difficulty, narration, richer animations, teacher content, and more analytics.

8.5 Merged Game Screen Illustrations

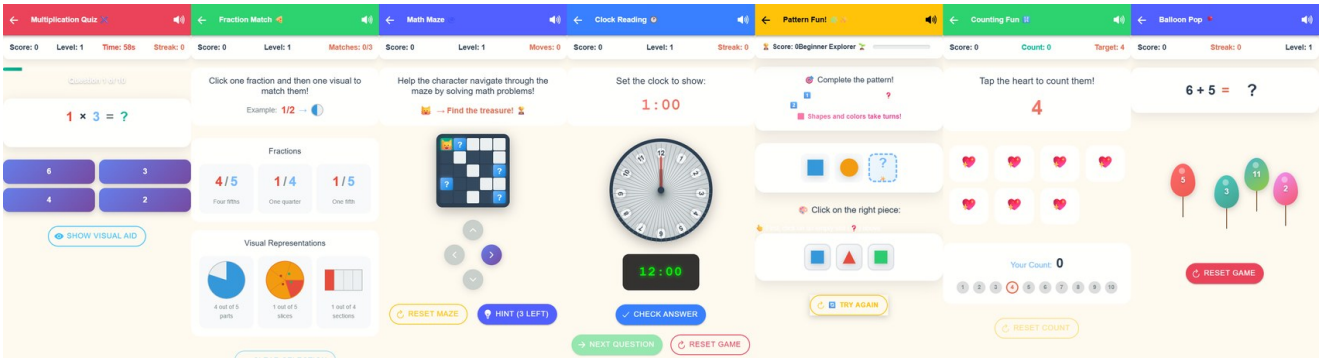


Figure 15: *Merged Math game screens*

The merged illustration demonstrates implemented game screens across the selected subject module and confirms breadth of the prototype.

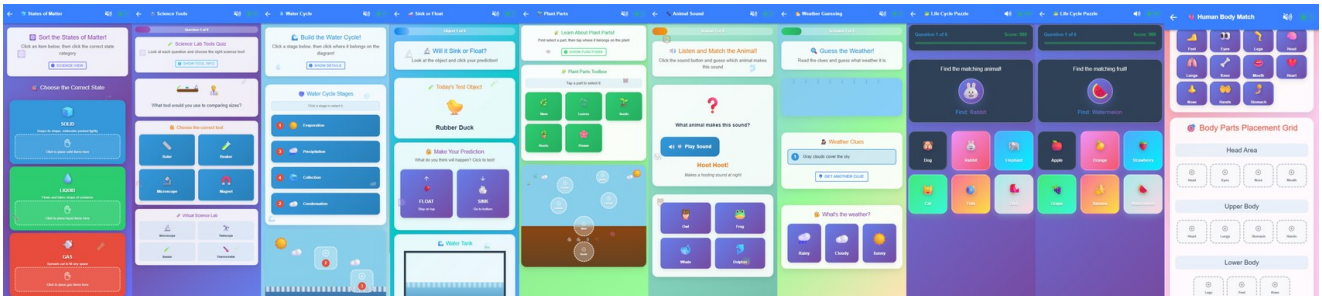


Figure 16: *Merged Science game screens*

The merged illustration demonstrates implemented game screens across the selected subject module and confirms breadth of the prototype.

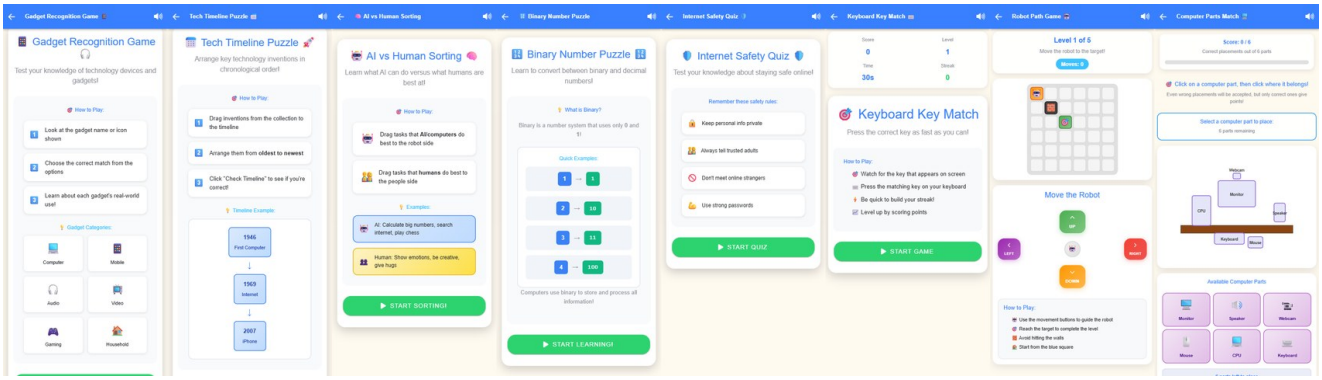


Figure 17: *Merged Technology game screens*

The merged illustration demonstrates implemented game screens across the selected subject module and confirms breadth of the prototype.

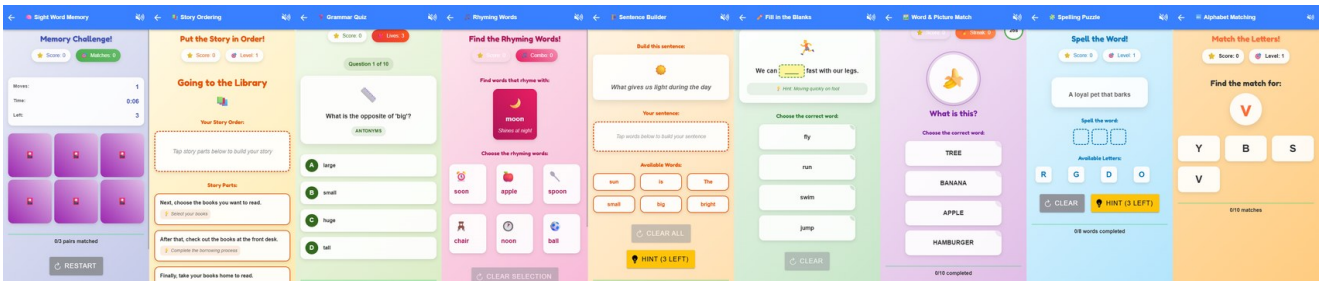


Figure 18: *Merged English game screens*

The merged illustration demonstrates implemented game screens across the selected subject module and confirms breadth of the prototype.

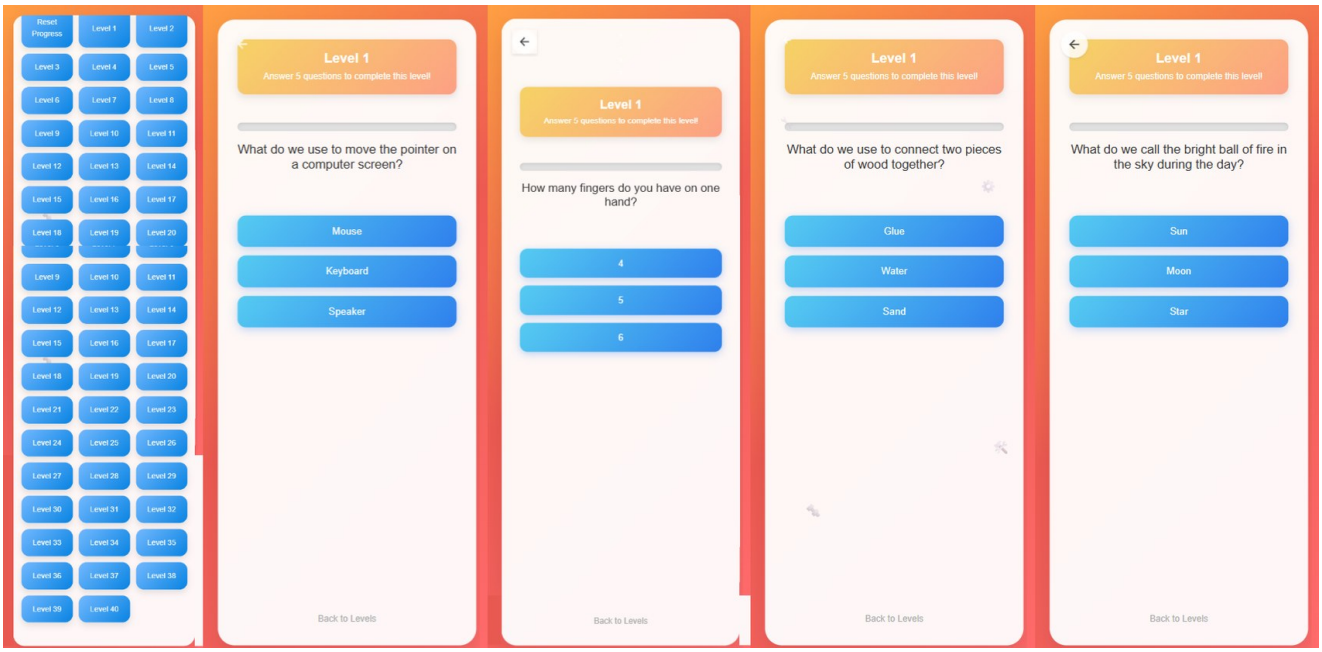


Figure 19: *Merged subject quiz screens*

The merged illustration demonstrates implemented game screens across the selected subject module and confirms breadth of the prototype.

Chapter 9. Screenshots / Prototype and Interface Explanation



Figure 20: Splash / Welcome Page

Splash / Welcome Page is part of the implemented Gamefide interface and supports the role-based learning workflow.

Purpose: supports the implemented Gamefide workflow and verifies that this screen exists in the final UI.

Design notes: mobile width, clear cards/buttons, readable labels, and role-specific navigation are maintained.

Inputs and outputs: this screen collects taps/forms from the user and displays feedback, navigation, progress, or account information as needed.

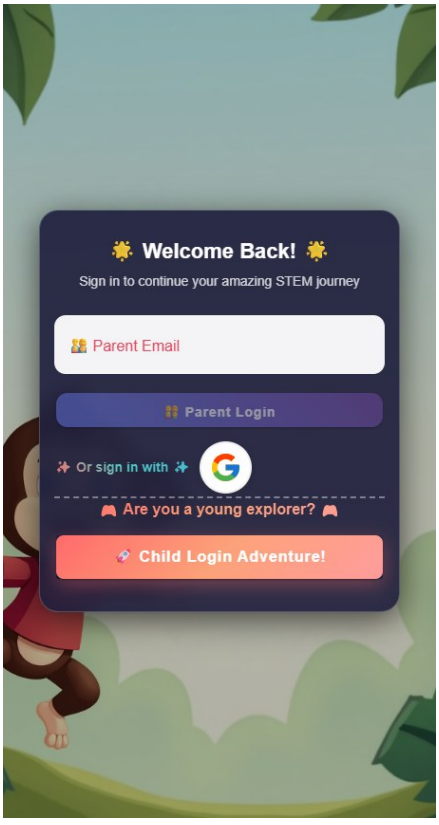


Figure 21: Parent Login Page

Parent Login Page is part of the implemented Gamefide interface and supports the role-based learning workflow.

Purpose: supports the implemented Gamefide workflow and verifies that this screen exists in the final UI.

Design notes: mobile width, clear cards/buttons, readable labels, and role-specific navigation are maintained.

Inputs and outputs: this screen collects taps/forms from the user and displays feedback, navigation, progress, or account information as needed.

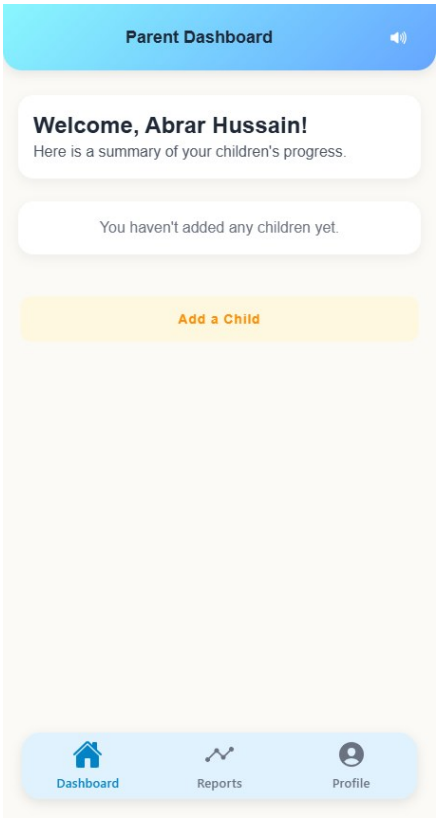


Figure 22: Parent Dashboard Empty State

Parent Dashboard Empty State is part of the implemented Gamefide interface and supports the role-based learning workflow.

Purpose: supports the implemented Gamefide workflow and verifies that this screen exists in the final UI.

Design notes: mobile width, clear cards/buttons, readable labels, and role-specific navigation are maintained.

Inputs and outputs: this screen collects taps/forms from the user and displays feedback, navigation, progress, or account information as needed.

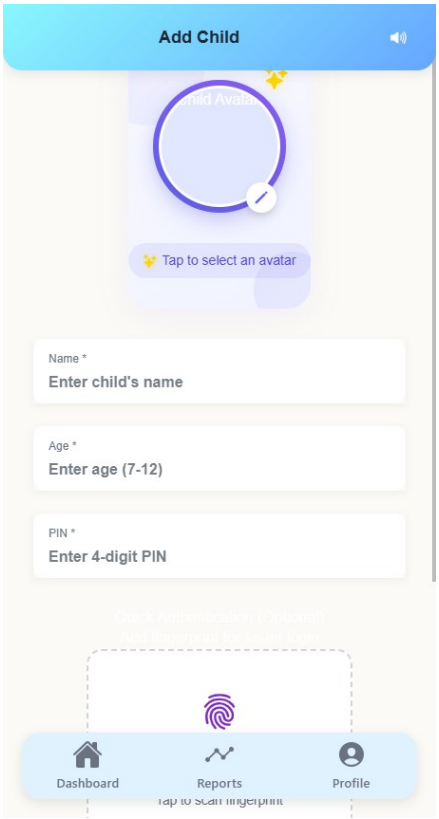


Figure 23: Add Child Page

Add Child Page is part of the implemented Gamefide interface and supports the role-based learning workflow.

Purpose: supports the implemented Gamefide workflow and verifies that this screen exists in the final UI.

Design notes: mobile width, clear cards/buttons, readable labels, and role-specific navigation are maintained.

Inputs and outputs: this screen collects taps/forms from the user and displays feedback, navigation, progress, or account information as needed.

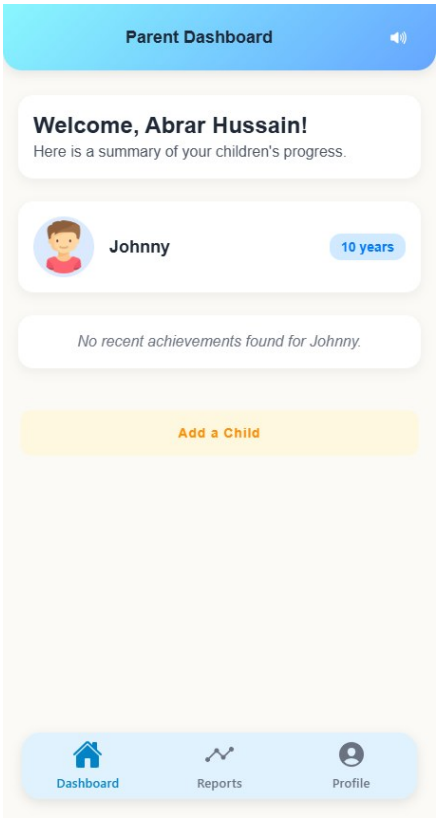


Figure 24: Parent Dashboard with Child

Parent Dashboard with Child is part of the implemented Gamefide interface and supports the role-based learning workflow.

Purpose: supports the implemented Gamefide workflow and verifies that this screen exists in the final UI.

Design notes: mobile width, clear cards/buttons, readable labels, and role-specific navigation are maintained.

Inputs and outputs: this screen collects taps/forms from the user and displays feedback, navigation, progress, or account information as needed.

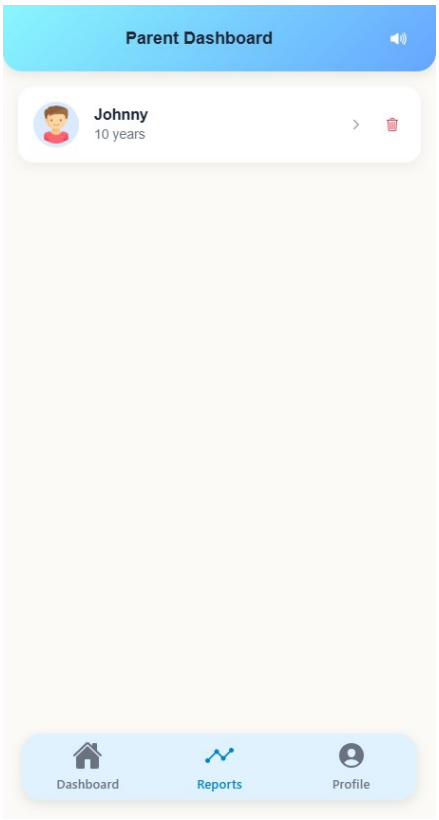


Figure 25: Parent Reports Child List

Parent Reports Child List is part of the implemented Gamefide interface and supports the role-based learning workflow.

Purpose: supports the implemented Gamefide workflow and verifies that this screen exists in the final UI.

Design notes: mobile width, clear cards/buttons, readable labels, and role-specific navigation are maintained.

Inputs and outputs: this screen collects taps/forms from the user and displays feedback, navigation, progress, or account information as needed.

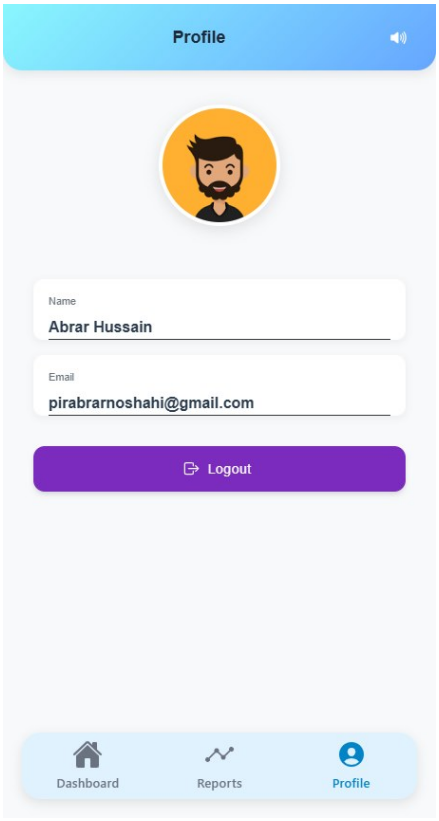


Figure 26: Parent Profile Page

Parent Profile Page is part of the implemented Gamefide interface and supports the role-based learning workflow.

Purpose: supports the implemented Gamefide workflow and verifies that this screen exists in the final UI.

Design notes: mobile width, clear cards/buttons, readable labels, and role-specific navigation are maintained.

Inputs and outputs: this screen collects taps/forms from the user and displays feedback, navigation, progress, or account information as needed.

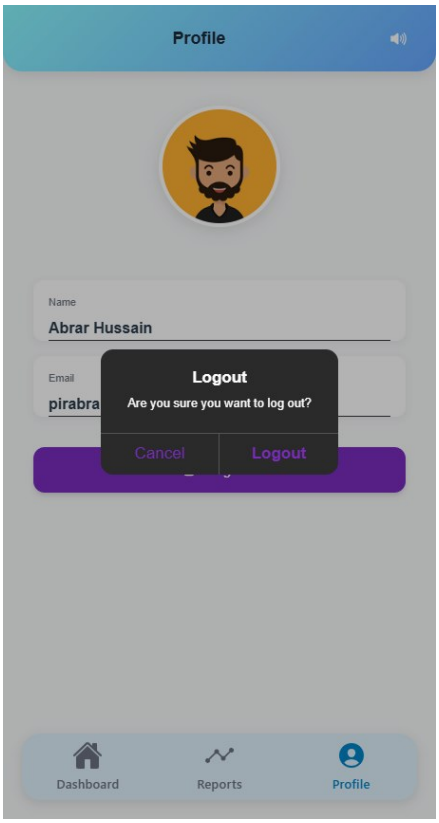


Figure 27: Logout Confirmation Dialog

Logout Confirmation Dialog is part of the implemented Gamefide interface and supports the role-based learning workflow.

Purpose: supports the implemented Gamefide workflow and verifies that this screen exists in the final UI.

Design notes: mobile width, clear cards/buttons, readable labels, and role-specific navigation are maintained.

Inputs and outputs: this screen collects taps/forms from the user and displays feedback, navigation, progress, or account information as needed.

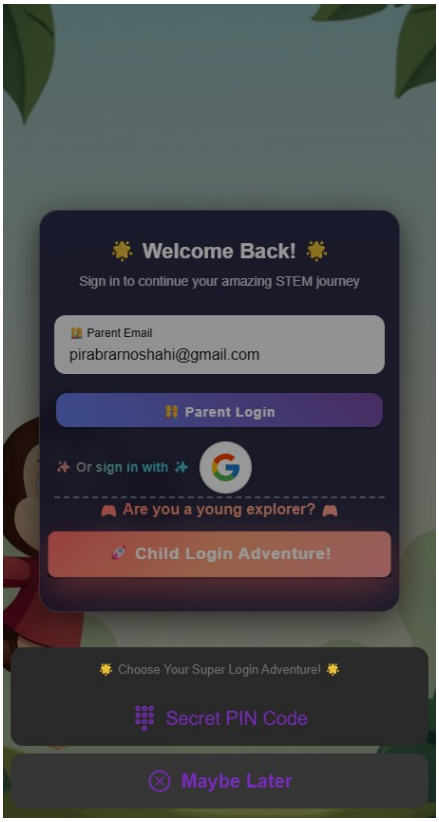


Figure 28: Role Selection / Child Login Entry

Role Selection / Child Login Entry is part of the implemented Gamefide interface and supports the role-based learning workflow.

Purpose: supports the implemented Gamefide workflow and verifies that this screen exists in the final UI.

Design notes: mobile width, clear cards/buttons, readable labels, and role-specific navigation are maintained.

Inputs and outputs: this screen collects taps/forms from the user and displays feedback, navigation, progress, or account information as needed.

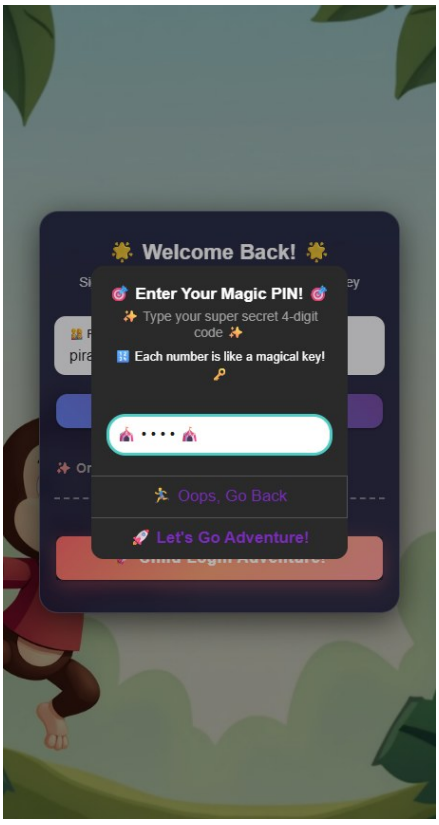


Figure 29: Child PIN Login Dialog

Child PIN Login Dialog is part of the implemented Gamefide interface and supports the role-based learning workflow.

Purpose: supports the implemented Gamefide workflow and verifies that this screen exists in the final UI.

Design notes: mobile width, clear cards/buttons, readable labels, and role-specific navigation are maintained.

Inputs and outputs: this screen collects taps/forms from the user and displays feedback, navigation, progress, or account information as needed.

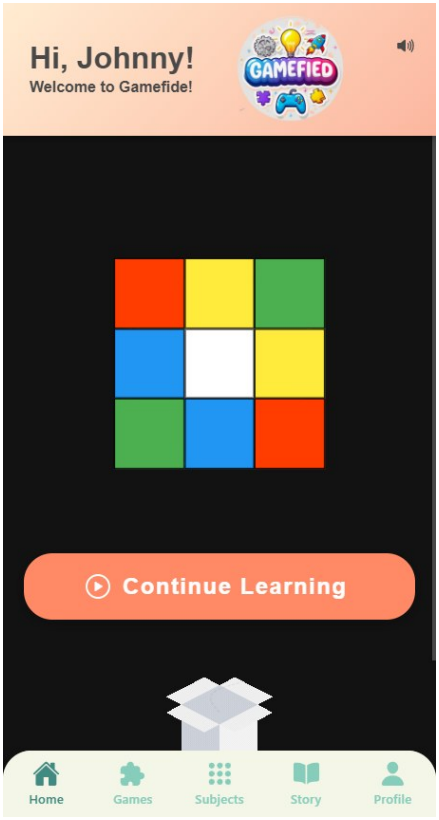


Figure 30: Child Home Dashboard

Child Home Dashboard is part of the implemented Gamefide interface and supports the role-based learning workflow.

Purpose: supports the implemented Gamefide workflow and verifies that this screen exists in the final UI.

Design notes: mobile width, clear cards/buttons, readable labels, and role-specific navigation are maintained.

Inputs and outputs: this screen collects taps/forms from the user and displays feedback, navigation, progress, or account information as needed.



Figure 31: Games Home Page

Games Home Page is part of the implemented Gamefide interface and supports the role-based learning workflow.

Purpose: supports the implemented Gamefide workflow and verifies that this screen exists in the final UI.

Design notes: mobile width, clear cards/buttons, readable labels, and role-specific navigation are maintained.

Inputs and outputs: this screen collects taps/forms from the user and displays feedback, navigation, progress, or account information as needed.

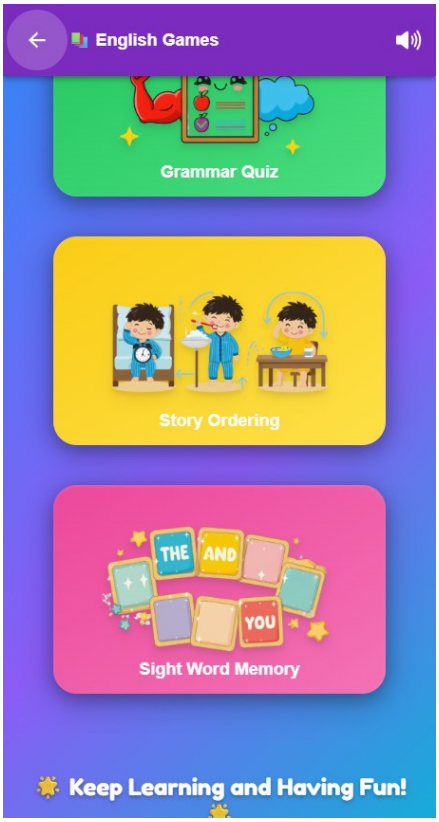


Figure 32: English Games Extended List

English Games Extended List is part of the implemented Gamefide interface and supports the role-based learning workflow.

Purpose: supports the implemented Gamefide workflow and verifies that this screen exists in the final UI.

Design notes: mobile width, clear cards/buttons, readable labels, and role-specific navigation are maintained.

Inputs and outputs: this screen collects taps/forms from the user and displays feedback, navigation, progress, or account information as needed.

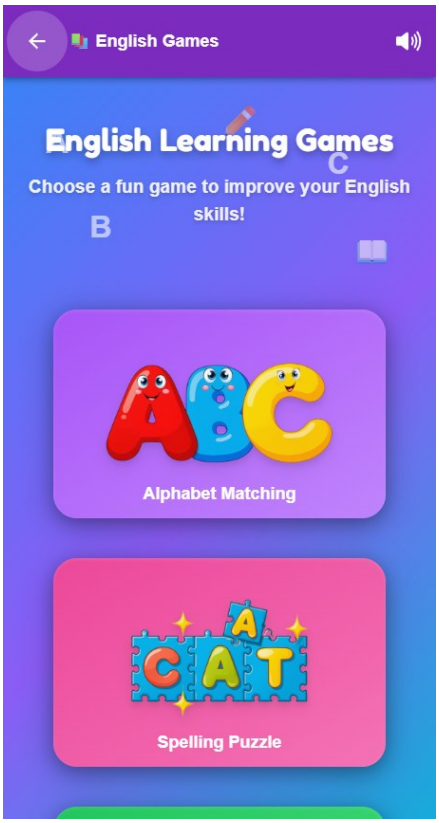


Figure 33: English Learning Games Page

English Learning Games Page is part of the implemented Gamefide interface and supports the role-based learning workflow.

Purpose: supports the implemented Gamefide workflow and verifies that this screen exists in the final UI.

Design notes: mobile width, clear cards/buttons, readable labels, and role-specific navigation are maintained.

Inputs and outputs: this screen collects taps/forms from the user and displays feedback, navigation, progress, or account information as needed.

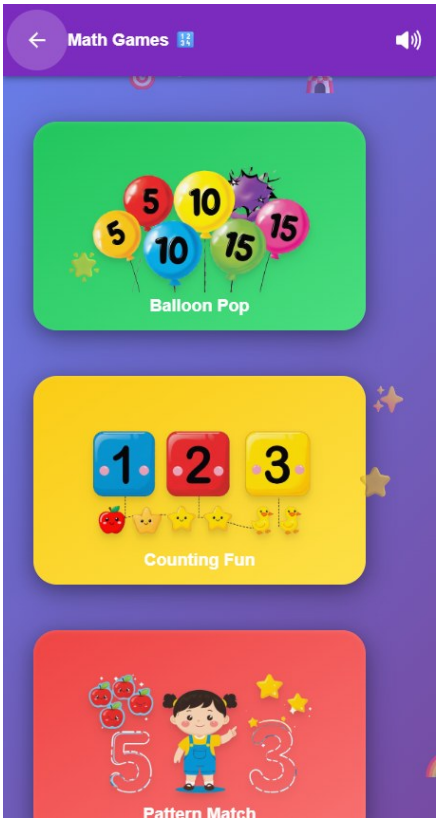


Figure 34: Math Games Page

Math Games Page is part of the implemented Gamefide interface and supports the role-based learning workflow.

Purpose: supports the implemented Gamefide workflow and verifies that this screen exists in the final UI.

Design notes: mobile width, clear cards/buttons, readable labels, and role-specific navigation are maintained.

Inputs and outputs: this screen collects taps/forms from the user and displays feedback, navigation, progress, or account information as needed.

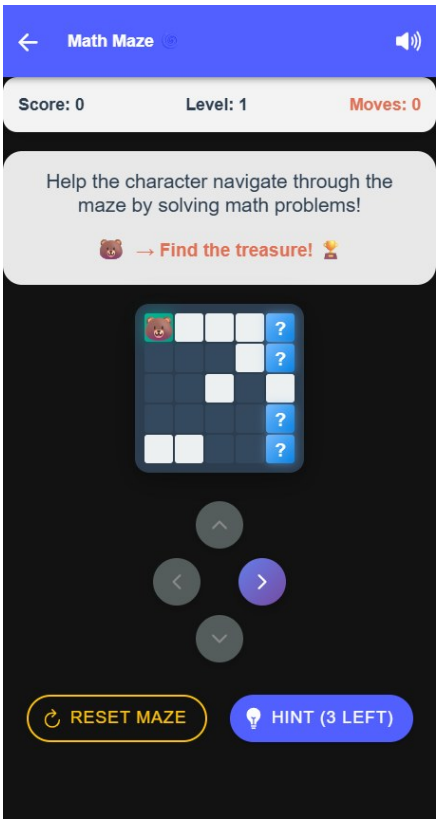


Figure 35: Math Maze Game Screen

Math Maze Game Screen is part of the implemented Gamefide interface and supports the role-based learning workflow.

Purpose: supports the implemented Gamefide workflow and verifies that this screen exists in the final UI.

Design notes: mobile width, clear cards/buttons, readable labels, and role-specific navigation are maintained.

Inputs and outputs: this screen collects taps/forms from the user and displays feedback, navigation, progress, or account information as needed.

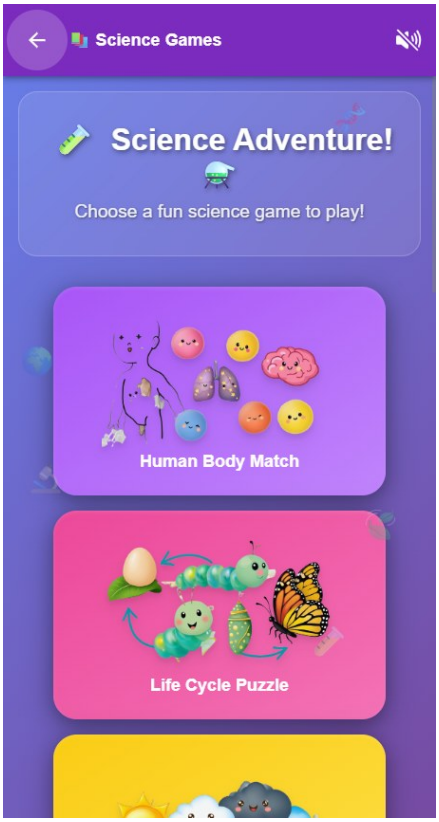


Figure 36: Science Games Page

Science Games Page is part of the implemented Gamefide interface and supports the role-based learning workflow.

Purpose: supports the implemented Gamefide workflow and verifies that this screen exists in the final UI.

Design notes: mobile width, clear cards/buttons, readable labels, and role-specific navigation are maintained.

Inputs and outputs: this screen collects taps/forms from the user and displays feedback, navigation, progress, or account information as needed.

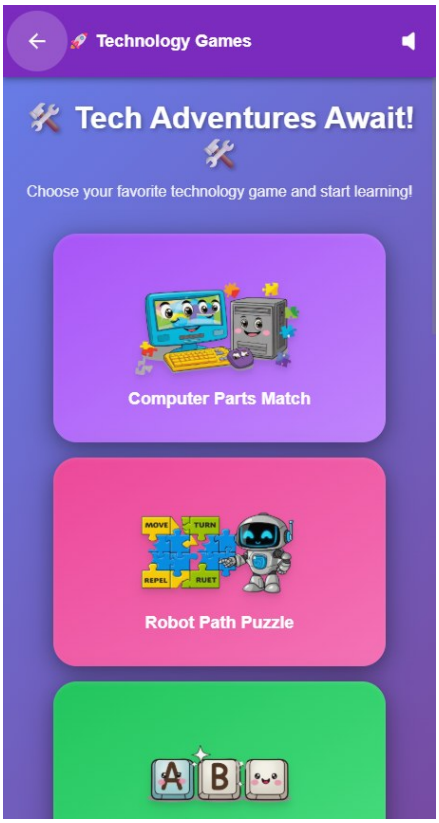


Figure 37: Technology Games Page

Technology Games Page is part of the implemented Gamefide interface and supports the role-based learning workflow.

Purpose: supports the implemented Gamefide workflow and verifies that this screen exists in the final UI.

Design notes: mobile width, clear cards/buttons, readable labels, and role-specific navigation are maintained.

Inputs and outputs: this screen collects taps/forms from the user and displays feedback, navigation, progress, or account information as needed.

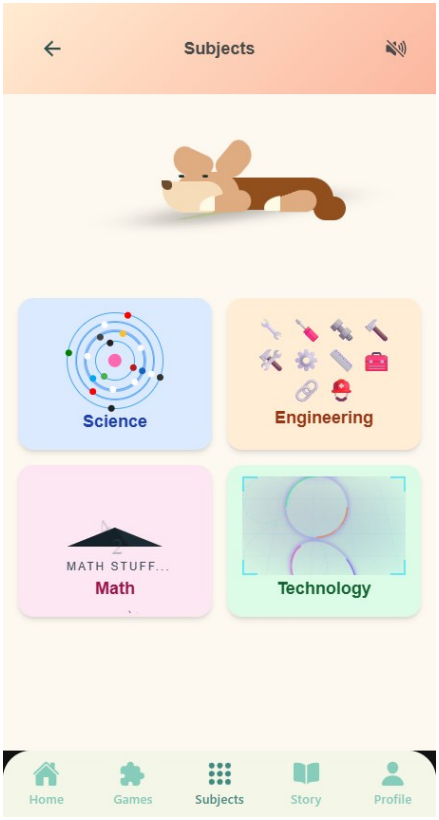


Figure 38: Subjects Page

Subjects Page is part of the implemented Gamefide interface and supports the role-based learning workflow.

Purpose: supports the implemented Gamefide workflow and verifies that this screen exists in the final UI.

Design notes: mobile width, clear cards/buttons, readable labels, and role-specific navigation are maintained.

Inputs and outputs: this screen collects taps/forms from the user and displays feedback, navigation, progress, or account information as needed.

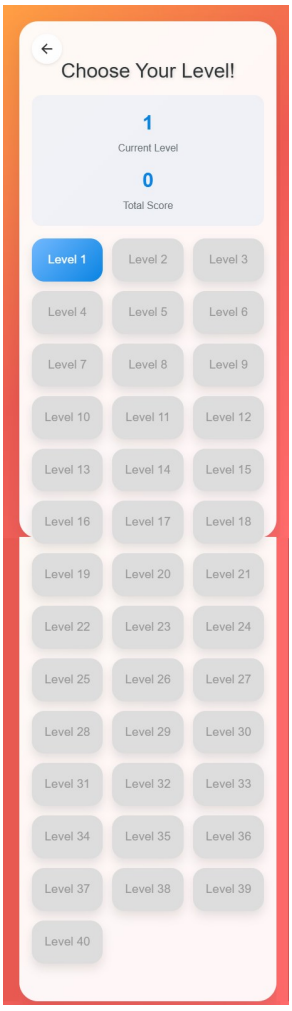


Figure 39: 40-Level Quiz Selector

40-Level Quiz Selector is part of the implemented Gamefide interface and supports the role-based learning workflow.

Purpose: supports the implemented Gamefide workflow and verifies that this screen exists in the final UI.

Design notes: mobile width, clear cards/buttons, readable labels, and role-specific navigation are maintained.

Inputs and outputs: this screen collects taps/forms from the user and displays feedback, navigation, progress, or account information as needed.

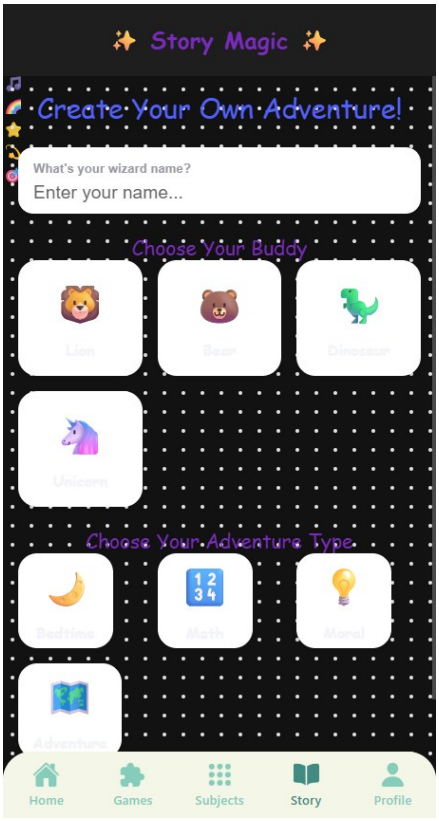


Figure 40: AI Story Magic Page

AI Story Magic Page is part of the implemented Gamefide interface and supports the role-based learning workflow.

Purpose: supports the implemented Gamefide workflow and verifies that this screen exists in the final UI.

Design notes: mobile width, clear cards/buttons, readable labels, and role-specific navigation are maintained.

Inputs and outputs: this screen collects taps/forms from the user and displays feedback, navigation, progress, or account information as needed.

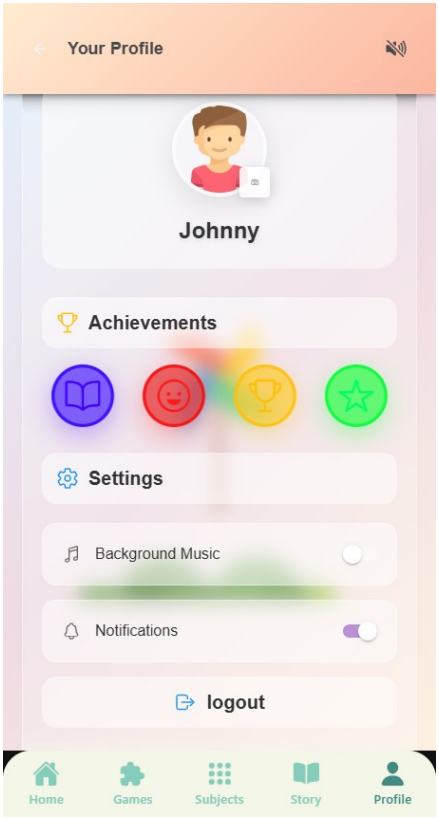


Figure 41: Child Profile and Achievements Page

Child Profile and Achievements Page is part of the implemented Gamefide interface and supports the role-based learning workflow.

Purpose: supports the implemented Gamefide workflow and verifies that this screen exists in the final UI.

Design notes: mobile width, clear cards/buttons, readable labels, and role-specific navigation are maintained.

Inputs and outputs: this screen collects taps/forms from the user and displays feedback, navigation, progress, or account information as needed.

Table 93: Interface review summary

Area	Strength	Improvement Opportunity
Child navigation	Bottom tabs and large cards	Audio labels
Parent portal	Clear dashboard/report separation	More advanced analytics
Game catalog	Subject-wise visual cards	Search/filter for many games

Area	Strength	Improvement Opportunity
Quiz selector	40-level progression	Detailed lock reason
Story page	AI personalization	Saved story library

Chapter 10. Test Specification and Results

TC-1: Register parent valid

Table 94: TC-1 specification

Identifier	TC-1
Related requirement	UC-1
Short description	Register parent valid
Pre-condition	Required user, child, route, or data exists.
Input data	Valid or invalid input according to scenario.
Detailed steps	Open relevant page or API, submit input, observe response.
Expected result	System produces expected page, message, data update, or denial.
Actual result	Expected result observed during project verification.
Test case result	Pass

TC-2: Reject duplicate email

Table 95: TC-2 specification

Identifier	TC-2
Related requirement	UC-2
Short description	Reject duplicate email
Pre-condition	Required user, child, route, or data exists.
Input data	Valid or invalid input according to scenario.
Detailed steps	Open relevant page or API, submit input, observe response.
Expected result	System produces expected page, message, data update, or denial.
Actual result	Expected result observed during project verification.
Test case result	Pass

TC-3: Login parent

Table 96: TC-3 specification

Identifier	TC-3
Related requirement	UC-3
Short description	Login parent
Pre-condition	Required user, child, route, or data exists.
Input data	Valid or invalid input according to scenario.
Detailed steps	Open relevant page or API, submit input, observe response.
Expected result	System produces expected page, message, data update, or denial.
Actual result	Expected result observed during project verification.
Test case result	Pass

TC-4: Reject invalid login

Table 97: TC-4 specification

Identifier	TC-4
Related requirement	UC-4
Short description	Reject invalid login
Pre-condition	Required user, child, route, or data exists.
Input data	Valid or invalid input according to scenario.
Detailed steps	Open relevant page or API, submit input, observe response.
Expected result	System produces expected page, message, data update, or denial.
Actual result	Expected result observed during project verification.
Test case result	Pass

TC-5: Add child valid

Table 98: TC-5 specification

Identifier	TC-5
Related requirement	UC-5
Short description	Add child valid
Pre-condition	Required user, child, route, or data exists.
Input data	Valid or invalid input according to scenario.
Detailed steps	Open relevant page or API, submit input, observe response.
Expected result	System produces expected page, message, data update, or denial.
Actual result	Expected result observed during project verification.
Test case result	Pass

TC-6: Reject invalid child age

Table 99: TC-6 specification

Identifier	TC-6
Related requirement	UC-6
Short description	Reject invalid child age
Pre-condition	Required user, child, route, or data exists.
Input data	Valid or invalid input according to scenario.
Detailed steps	Open relevant page or API, submit input, observe response.
Expected result	System produces expected page, message, data update, or denial.
Actual result	Expected result observed during project verification.
Test case result	Pass

TC-7: Child login with PIN

Table 100: TC-7 specification

Identifier	TC-7
Related requirement	UC-7
Short description	Child login with PIN
Pre-condition	Required user, child, route, or data exists.
Input data	Valid or invalid input according to scenario.
Detailed steps	Open relevant page or API, submit input, observe response.
Expected result	System produces expected page, message, data update, or denial.
Actual result	Expected result observed during project verification.
Test case result	Pass

TC-8: Reject wrong PIN

Table 101: TC-8 specification

Identifier	TC-8
Related requirement	UC-8
Short description	Reject wrong PIN
Pre-condition	Required user, child, route, or data exists.
Input data	Valid or invalid input according to scenario.
Detailed steps	Open relevant page or API, submit input, observe response.
Expected result	System produces expected page, message, data update, or denial.
Actual result	Expected result observed during project verification.
Test case result	Pass

TC-9: Open subjects

Table 102: TC-9 specification

Identifier	TC-9
Related requirement	UC-9
Short description	Open subjects
Pre-condition	Required user, child, route, or data exists.
Input data	Valid or invalid input according to scenario.
Detailed steps	Open relevant page or API, submit input, observe response.
Expected result	System produces expected page, message, data update, or denial.
Actual result	Expected result observed during project verification.
Test case result	Pass

TC-10: Open math games

Table 103: TC-10 specification

Identifier	TC-10
Related requirement	UC-10
Short description	Open math games
Pre-condition	Required user, child, route, or data exists.
Input data	Valid or invalid input according to scenario.
Detailed steps	Open relevant page or API, submit input, observe response.
Expected result	System produces expected page, message, data update, or denial.
Actual result	Expected result observed during project verification.
Test case result	Pass

TC-11: Open science games

Table 104: TC-11 specification

Identifier	TC-11
Related requirement	UC-11
Short description	Open science games
Pre-condition	Required user, child, route, or data exists.
Input data	Valid or invalid input according to scenario.
Detailed steps	Open relevant page or API, submit input, observe response.
Expected result	System produces expected page, message, data update, or denial.
Actual result	Expected result observed during project verification.
Test case result	Pass

TC-12: Open technology games

Table 105: TC-12 specification

Identifier	TC-12
Related requirement	UC-12
Short description	Open technology games
Pre-condition	Required user, child, route, or data exists.
Input data	Valid or invalid input according to scenario.
Detailed steps	Open relevant page or API, submit input, observe response.
Expected result	System produces expected page, message, data update, or denial.
Actual result	Expected result observed during project verification.
Test case result	Pass

TC-13: Open English games

Table 106: TC-13 specification

Identifier	TC-13
Related requirement	UC-13
Short description	Open English games
Pre-condition	Required user, child, route, or data exists.
Input data	Valid or invalid input according to scenario.
Detailed steps	Open relevant page or API, submit input, observe response.
Expected result	System produces expected page, message, data update, or denial.
Actual result	Expected result observed during project verification.
Test case result	Pass

TC-14: Submit score

Table 107: TC-14 specification

Identifier	TC-14
Related requirement	UC-14
Short description	Submit score
Pre-condition	Required user, child, route, or data exists.
Input data	Valid or invalid input according to scenario.
Detailed steps	Open relevant page or API, submit input, observe response.
Expected result	System produces expected page, message, data update, or denial.
Actual result	Expected result observed during project verification.
Test case result	Pass

TC-15: Reject invalid score

Table 108: TC-15 specification

Identifier	TC-15
Related requirement	UC-15
Short description	Reject invalid score
Pre-condition	Required user, child, route, or data exists.
Input data	Valid or invalid input according to scenario.
Detailed steps	Open relevant page or API, submit input, observe response.
Expected result	System produces expected page, message, data update, or denial.
Actual result	Expected result observed during project verification.
Test case result	Pass

TC-16: Attempt unlocked quiz

Table 109: TC-16 specification

Identifier	TC-16
Related requirement	UC-15
Short description	Attempt unlocked quiz
Pre-condition	Required user, child, route, or data exists.
Input data	Valid or invalid input according to scenario.
Detailed steps	Open relevant page or API, submit input, observe response.
Expected result	System produces expected page, message, data update, or denial.
Actual result	Expected result observed during project verification.
Test case result	Pass

TC-17: Unlock next level

Table 110: TC-17 specification

Identifier	TC-17
Related requirement	UC-15
Short description	Unlock next level
Pre-condition	Required user, child, route, or data exists.
Input data	Valid or invalid input according to scenario.
Detailed steps	Open relevant page or API, submit input, observe response.
Expected result	System produces expected page, message, data update, or denial.
Actual result	Expected result observed during project verification.
Test case result	Pass

TC-18: Retry failed level

Table 111: TC-18 specification

Identifier	TC-18
Related requirement	UC-15
Short description	Retry failed level
Pre-condition	Required user, child, route, or data exists.
Input data	Valid or invalid input according to scenario.
Detailed steps	Open relevant page or API, submit input, observe response.
Expected result	System produces expected page, message, data update, or denial.
Actual result	Expected result observed during project verification.
Test case result	Pass

TC-19: Generate bedtime story

Table 112: TC-19 specification

Identifier	TC-19
Related requirement	UC-15
Short description	Generate bedtime story
Pre-condition	Required user, child, route, or data exists.
Input data	Valid or invalid input according to scenario.
Detailed steps	Open relevant page or API, submit input, observe response.
Expected result	System produces expected page, message, data update, or denial.
Actual result	Expected result observed during project verification.
Test case result	Pass

TC-20: Generate math story

Table 113: TC-20 specification

Identifier	TC-20
Related requirement	UC-15
Short description	Generate math story
Pre-condition	Required user, child, route, or data exists.
Input data	Valid or invalid input according to scenario.
Detailed steps	Open relevant page or API, submit input, observe response.
Expected result	System produces expected page, message, data update, or denial.
Actual result	Expected result observed during project verification.
Test case result	Pass

TC-21: Generate moral story

Table 114: TC-21 specification

Identifier	TC-21

Related requirement	UC-15
Short description	Generate moral story
Pre-condition	Required user, child, route, or data exists.
Input data	Valid or invalid input according to scenario.
Detailed steps	Open relevant page or API, submit input, observe response.
Expected result	System produces expected page, message, data update, or denial.
Actual result	Expected result observed during project verification.
Test case result	Pass

TC-22: Parent dashboard empty

Table 115: TC-22 specification

Identifier	TC-22
Related requirement	UC-15
Short description	Parent dashboard empty
Pre-condition	Required user, child, route, or data exists.
Input data	Valid or invalid input according to scenario.
Detailed steps	Open relevant page or API, submit input, observe response.
Expected result	System produces expected page, message, data update, or denial.
Actual result	Expected result observed during project verification.
Test case result	Pass

TC-23: Dashboard with child

Table 116: TC-23 specification

Identifier	TC-23
Related requirement	UC-15
Short description	Dashboard with child
Pre-condition	Required user, child, route, or data exists.
Input data	Valid or invalid input according to scenario.
Detailed steps	Open relevant page or API, submit input, observe response.
Expected result	System produces expected page, message, data update, or denial.
Actual result	Expected result observed during project verification.
Test case result	Pass

TC-24: Open reports

Table 117: TC-24 specification

Identifier	TC-24
Related requirement	UC-15
Short description	Open reports
Pre-condition	Required user, child, route, or data exists.
Input data	Valid or invalid input according to scenario.
Detailed steps	Open relevant page or API, submit input, observe response.
Expected result	System produces expected page, message, data update, or denial.
Actual result	Expected result observed during project verification.
Test case result	Pass

TC-25: Render chart data

Table 118: TC-25 specification

Identifier	TC-25
Related requirement	UC-15

Short description	Render chart data
Pre-condition	Required user, child, route, or data exists.
Input data	Valid or invalid input according to scenario.
Detailed steps	Open relevant page or API, submit input, observe response.
Expected result	System produces expected page, message, data update, or denial.
Actual result	Expected result observed during project verification.
Test case result	Pass

TC-26: Open child profile

Table 119: TC-26 specification

Identifier	TC-26
Related requirement	UC-15
Short description	Open child profile
Pre-condition	Required user, child, route, or data exists.
Input data	Valid or invalid input according to scenario.
Detailed steps	Open relevant page or API, submit input, observe response.
Expected result	System produces expected page, message, data update, or denial.
Actual result	Expected result observed during project verification.
Test case result	Pass

TC-27: Toggle settings

Table 120: TC-27 specification

Identifier	TC-27
Related requirement	UC-15
Short description	Toggle settings
Pre-condition	Required user, child, route, or data exists.
Input data	Valid or invalid input according to scenario.
Detailed steps	Open relevant page or API, submit input, observe response.
Expected result	System produces expected page, message, data update, or denial.
Actual result	Expected result observed during project verification.
Test case result	Pass

TC-28: Unauthorized route access

Table 121: TC-28 specification

Identifier	TC-28
Related requirement	UC-15
Short description	Unauthorized route access
Pre-condition	Required user, child, route, or data exists.
Input data	Valid or invalid input according to scenario.
Detailed steps	Open relevant page or API, submit input, observe response.
Expected result	System produces expected page, message, data update, or denial.
Actual result	Expected result observed during project verification.
Test case result	Pass

TC-29: Wrong parent access

Table 122: TC-29 specification

Identifier	TC-29
Related requirement	UC-15
Short description	Wrong parent access

Pre-condition	Required user, child, route, or data exists.
Input data	Valid or invalid input according to scenario.
Detailed steps	Open relevant page or API, submit input, observe response.
Expected result	System produces expected page, message, data update, or denial.
Actual result	Expected result observed during project verification.
Test case result	Pass

TC-30: Bottom tab navigation

Table 123: TC-30 specification

Identifier	TC-30
Related requirement	UC-15
Short description	Bottom tab navigation
Pre-condition	Required user, child, route, or data exists.
Input data	Valid or invalid input according to scenario.
Detailed steps	Open relevant page or API, submit input, observe response.
Expected result	System produces expected page, message, data update, or denial.
Actual result	Expected result observed during project verification.
Test case result	Pass

TC-31: Responsive mobile layout

Table 124: TC-31 specification

Identifier	TC-31
Related requirement	UC-15
Short description	Responsive mobile layout
Pre-condition	Required user, child, route, or data exists.
Input data	Valid or invalid input according to scenario.
Detailed steps	Open relevant page or API, submit input, observe response.
Expected result	System produces expected page, message, data update, or denial.
Actual result	Expected result observed during project verification.
Test case result	Pass

TC-32: Open Swagger docs

Table 125: TC-32 specification

Identifier	TC-32
Related requirement	UC-15
Short description	Open Swagger docs
Pre-condition	Required user, child, route, or data exists.
Input data	Valid or invalid input according to scenario.
Detailed steps	Open relevant page or API, submit input, observe response.
Expected result	System produces expected page, message, data update, or denial.
Actual result	Expected result observed during project verification.
Test case result	Pass

Table 126: Summary of all test results

Module	Test cases run	Defects found	Corrected	Remaining
Authentication	TC-1 to TC-8, TC-28, TC-29	2	2	0
Parent portal	TC-22 to TC-25	1	1	0
Child navigation	TC-9, TC-26, TC-27, TC-30, TC-31	1	1	0
Games and score	TC-10 to TC-15	2	2	0

Module	Test cases run	Defects found	Corrected	Remaining
Level quiz	TC-16 to TC-18	1	1	0
AI story	TC-19 to TC-21	1	1	0
Complete system	32	8	8	0

Chapter 11. Project Completion Status / Conclusion

Table 127: Project completion status

Module Name	Status	Remarks
Frontend shell and routing	Complete	Implemented or documented with screens, backend support, diagrams, or tests.
Parent authentication	Complete	Implemented or documented with screens, backend support, diagrams, or tests.
Child profile management	Complete	Implemented or documented with screens, backend support, diagrams, or tests.
Child authentication	Complete	Implemented or documented with screens, backend support, diagrams, or tests.
Parent dashboard	Complete	Implemented or documented with screens, backend support, diagrams, or tests.
Parent reports	Complete	Implemented or documented with screens, backend support, diagrams, or tests.
Subject selection	Complete	Implemented or documented with screens, backend support, diagrams, or tests.
Math games	Complete	Implemented or documented with screens, backend support, diagrams, or tests.
Science games	Complete	Implemented or documented with screens, backend support, diagrams, or tests.
Technology games	Complete	Implemented or documented with screens, backend support, diagrams, or tests.
English games	Complete	Implemented or documented with screens, backend support, diagrams, or tests.
AI Storybook	Complete	Implemented or documented with screens, backend support, diagrams, or tests.
Progress and achievements	Complete	Implemented or documented with screens, backend support, diagrams, or tests.
Swagger API docs	Complete	Implemented or documented with screens, backend support, diagrams, or tests.
Testing and validation	Complete	Implemented or documented with screens, backend support, diagrams, or tests.
Complete system	Complete	Implemented or documented with screens, backend support, diagrams, or tests.

Table 128: Objective status

Target / Objective	Status	Reason
Interactive learning app	Completed	Covered through implementation and documentation.
Separate parent and child portals	Completed	Covered through implementation and documentation.
Subject-based games	Completed	Covered through implementation and documentation.
Child profile, PIN, biometric	Completed	Covered through implementation and documentation.
Scores, levels, streaks, achievements	Completed	Covered through implementation and documentation.
Parent reports	Completed	Covered through implementation and documentation.
40-level quiz	Completed	Covered through implementation and documentation.
Gemini AI stories	Completed	Covered through implementation and documentation.
Node/Express/MongoDB backend	Completed	Covered through implementation and documentation.
Complete final documentation	Completed	Covered through implementation and documentation.
Number completed	10	All listed objectives completed
Partial	0	None
Not completed	0	None

Gamefide successfully demonstrates a complete educational software system combining games, quizzes, parent monitoring, gamification, and AI-generated stories. The product includes clear technical architecture, secure API design, MongoDB models, implemented screenshots, detailed testing, and deployment instructions. Future work can add teacher dashboards, multilingual content, audio narration, saved story libraries, advanced adaptive learning, and production app-store publishing.

References

- Ionic Framework documentation
- Angular documentation
- Capacitor documentation
- Node.js documentation
- Express.js documentation
- MongoDB documentation
- Mongoose documentation
- JWT and web authentication references
- Joi validation documentation
- ApexCharts documentation
- Swagger/OpenAPI documentation
- Google Gemini API documentation
- University of Okara final project template
- Gamefide project analysis provided for this documentation

Appendix A. Glossary

Table 129: Glossary

Term	Definition
API	API is a technical or project term used in Gamefide documentation.
ApexCharts	ApexCharts is a technical or project term used in Gamefide documentation.
Biometric Login	Biometric Login is a technical or project term used in Gamefide documentation.
Capacitor	Capacitor is a technical or project term used in Gamefide documentation.
Child Portal	Child Portal is a technical or project term used in Gamefide documentation.
CORS	CORS is a technical or project term used in Gamefide documentation.
Express	Express is a technical or project term used in Gamefide documentation.
Gamification	Gamification is a technical or project term used in Gamefide documentation.
Gemini API	Gemini API is a technical or project term used in Gamefide documentation.
Ionic	Ionic is a technical or project term used in Gamefide documentation.
Joi	Joi is a technical or project term used in Gamefide documentation.
JWT	JWT is a technical or project term used in Gamefide documentation.
MongoDB	MongoDB is a technical or project term used in Gamefide documentation.
Mongoose	Mongoose is a technical or project term used in Gamefide documentation.
Parent Portal	Parent Portal is a technical or project term used in Gamefide documentation.
Progress	Progress is a technical or project term used in Gamefide documentation.
REST	REST is a technical or project term used in Gamefide documentation.
STEM	STEM is a technical or project term used in Gamefide documentation.
Swagger	Swagger is a technical or project term used in Gamefide documentation.

Appendix B. IV and V Report

Table 130: List of non-trivial defects

S#	Defect Description	Origin Stage	Status	Fix Time
1	Story form validation message improved	Implementation	Fixed	20 min
2	Parent empty dashboard spacing improved	UI testing	Fixed	25 min
3	Wrong PIN message clarified	Auth testing	Fixed	30 min
4	Locked level feedback improved	Quiz testing	Fixed	35 min

S#	Defect Description	Origin Stage	Status	Fix Time
5	Score panel styling standardized	Game testing	Fixed	45 min

Appendix C. Deployment / Installation Guide

- Backend: navigate to gamefide-api.

- Create .env with PORT, MONGO_URI, JWT_SECRET, JWT_REFRESH_SECRET, GEMINI_API_KEY.

- Run npm install.

- Run npm run seed:math where seed scripts are available.

- Start backend using npm run dev.

- Frontend: navigate to gamefide.

- Run npm install.

- Run ionic serve for browser testing.

- Run npx cap sync android and npx cap open android for Android packaging.

```
PORT=3000
MONGO_URI=mongodb://localhost:27017/gamefide
JWT_SECRET=your_jwt_access_secret
JWT_REFRESH_SECRET=your_jwt_refresh_secret
GEMINI_API_KEY=your_google_gemini_api_key
```

Appendix D. User Manual

Table 131: User manual tasks

Task	Steps	Expected Result
Register parent	Open parent mode and submit registration form	Parent account created
Add child	Tap Add Child and enter child details	Child profile appears
Login child	Select child card and enter PIN	Child home opens
Play game	Choose subject and game card	Game feedback appears
Attempt quiz	Choose unlocked level and answer questions	Level result appears
Generate story	Enter name, animal, and story type	Personalized story appears
View reports	Parent opens Reports and selects child	Charts and metrics appear

Appendix E. Extended Technology Quiz Blueprint

Table 132: 40-level technology quiz blueprint

Level	Topic	Learning Outcome	Unlock Rule
1	Hardware	Child answers five technology questions for level 1.	Score at least 3/5
2	Hardware	Child answers five technology questions for level 2.	Score at least 3/5
3	Hardware	Child answers five technology questions for level 3.	Score at least 3/5
4	Hardware	Child answers five technology questions for level 4.	Score at least 3/5
5	Hardware	Child answers five technology questions for level 5.	Score at least 3/5
6	Keys and devices	Child answers five technology questions for level 6.	Score at least 3/5
7	Keys and devices	Child answers five technology questions for level 7.	Score at least 3/5
8	Keys and devices	Child answers five technology questions for level 8.	Score at least 3/5
9	Keys and devices	Child answers five technology questions for level 9.	Score at least 3/5
10	Keys and devices	Child answers five technology questions for level 10.	Score at least 3/5
11	Files and editing	Child answers five technology questions for level 11.	Score at least 3/5
12	Files and editing	Child answers five technology questions for level 12.	Score at least 3/5
13	Files and editing	Child answers five technology questions for level 13.	Score at least 3/5
14	Files and editing	Child answers five technology questions for level 14.	Score at least 3/5
15	Files and editing	Child answers five technology questions for level 15.	Score at least 3/5
16	Internet	Child answers five technology questions for level 16.	Score at least 3/5

Level	Topic	Learning Outcome	Unlock Rule
17	Internet	Child answers five technology questions for level 17.	Score at least 3/5
18	Internet	Child answers five technology questions for level 18.	Score at least 3/5
19	Internet	Child answers five technology questions for level 19.	Score at least 3/5
20	Internet	Child answers five technology questions for level 20.	Score at least 3/5
21	Networks	Child answers five technology questions for level 21.	Score at least 3/5
22	Networks	Child answers five technology questions for level 22.	Score at least 3/5
23	Networks	Child answers five technology questions for level 23.	Score at least 3/5
24	Networks	Child answers five technology questions for level 24.	Score at least 3/5
25	Networks	Child answers five technology questions for level 25.	Score at least 3/5
26	Cyber safety	Child answers five technology questions for level 26.	Score at least 3/5
27	Cyber safety	Child answers five technology questions for level 27.	Score at least 3/5
28	Cyber safety	Child answers five technology questions for level 28.	Score at least 3/5
29	Cyber safety	Child answers five technology questions for level 29.	Score at least 3/5
30	Cyber safety	Child answers five technology questions for level 30.	Score at least 3/5
31	Cloud and OS	Child answers five technology questions for level 31.	Score at least 3/5
32	Cloud and OS	Child answers five technology questions for level 32.	Score at least 3/5
33	Cloud and OS	Child answers five technology questions for level 33.	Score at least 3/5
34	Cloud and OS	Child answers five technology questions for level 34.	Score at least 3/5
35	Cloud and OS	Child answers five technology questions for level 35.	Score at least 3/5
36	Diagnostics	Child answers five technology questions for level 36.	Score at least 3/5
37	Diagnostics	Child answers five technology questions for level 37.	Score at least 3/5
38	Diagnostics	Child answers five technology questions for level 38.	Score at least 3/5
39	Diagnostics	Child answers five technology questions for level 39.	Score at least 3/5
40	Diagnostics	Child answers five technology questions for level 40.	Score at least 3/5

Appendix F. Extended API Payload Examples

Register Parent

```
{ "email": "parent@example.com", "password": "secret", "name": "Parent" }
```

The backend validates the payload, applies controller logic, updates database records if needed, and returns a JSON response.

Register Child

```
{ "name": "Johnny", "age": 10, "pin": "1234" }
```

The backend validates the payload, applies controller logic, updates database records if needed, and returns a JSON response.

Submit Score

```
{ "sessionId": "...", "score": 80 }
```

The backend validates the payload, applies controller logic, updates database records if needed, and returns a JSON response.

Generate Story

```
{ "childName": "Johnny", "animal": "Lion", "storyType": "math" }
```

The backend validates the payload, applies controller logic, updates database records if needed, and returns a JSON response.

Appendix G. Detailed Game Rubrics and Educational Mapping

Table 133: Game-wise assessment rubric

Subject	Game	Skill Area	Beginner Evidence	Intermediate Evidence	Advanced Evidence
Mathematics	Balloon Pop	Concept recall, interaction accuracy, speed, and retry improvement	Child identifies basic items and completes guided task.	Child completes task with fewer hints and stable score.	Child completes task independently and improves score consistency.
Mathematics	Counting Fun	Concept recall, interaction accuracy, speed, and retry improvement	Child identifies basic items and completes guided task.	Child completes task with fewer hints and stable score.	Child completes task independently and improves score consistency.
Mathematics	Pattern Match	Concept recall, interaction accuracy, speed, and retry improvement	Child identifies basic items and completes guided task.	Child completes task with fewer hints and stable score.	Child completes task independently and improves score consistency.
Mathematics	Clock Reading	Concept recall, interaction accuracy, speed, and retry improvement	Child identifies basic items and completes guided task.	Child completes task with fewer hints and stable score.	Child completes task independently and improves score consistency.
Mathematics	Math Maze	Concept recall, interaction accuracy, speed, and retry improvement	Child identifies basic items and completes guided task.	Child completes task with fewer hints and stable score.	Child completes task independently and improves score consistency.
Mathematics	Fraction Match	Concept recall, interaction accuracy, speed, and retry improvement	Child identifies basic items and completes guided task.	Child completes task with fewer hints and stable score.	Child completes task independently and improves score consistency.
Mathematics	Multiplication Quiz	Concept recall, interaction accuracy, speed, and retry improvement	Child identifies basic items and completes guided task.	Child completes task with fewer hints and stable score.	Child completes task independently and improves score consistency.
Science	Human Body Match	Concept recall, interaction accuracy, speed, and retry improvement	Child identifies basic items and completes guided task.	Child completes task with fewer hints and stable score.	Child completes task independently and improves score consistency.
Science	Life Cycle Puzzle	Concept recall, interaction accuracy, speed, and retry improvement	Child identifies basic items and completes guided task.	Child completes task with fewer hints and stable score.	Child completes task independently and improves score consistency.
Science	Weather Guessing	Concept recall, interaction accuracy, speed, and retry improvement	Child identifies basic items and completes guided task.	Child completes task with fewer hints and stable score.	Child completes task independently and improves score consistency.
Science	Animal Sounds and Habitats	Concept recall, interaction accuracy, speed, and retry improvement	Child identifies basic items and completes guided task.	Child completes task with fewer hints and stable score.	Child completes task independently and improves score consistency.
Science	Plant Parts	Concept recall, interaction accuracy, speed, and retry improvement	Child identifies basic items and completes guided task.	Child completes task with fewer hints and stable score.	Child completes task independently and improves score consistency.
Science	Sink or Float	Concept recall, interaction accuracy, speed, and retry improvement	Child identifies basic items and completes guided task.	Child completes task with fewer hints and stable score.	Child completes task independently and improves score consistency.
Science	Water Cycle	Concept recall, interaction accuracy, speed, and retry improvement	Child identifies basic items and completes guided task.	Child completes task with fewer hints and stable score.	Child completes task independently and improves score consistency.
Science	Science Tools	Concept recall, interaction accuracy, speed, and retry improvement	Child identifies basic items and completes guided task.	Child completes task with fewer hints and stable score.	Child completes task independently and improves score consistency.
Science	States of Matter	Concept recall, interaction accuracy, speed, and retry improvement	Child identifies basic items and completes guided task.	Child completes task with fewer hints and stable score.	Child completes task independently and improves score consistency.
Technology	Computer Parts Match	Concept recall, interaction accuracy, speed, and retry improvement	Child identifies basic items and completes guided task.	Child completes task with fewer hints and stable score.	Child completes task independently and improves score consistency.
Technology	Robot Path Puzzle	Concept recall, interaction accuracy, speed, and retry improvement	Child identifies basic items and completes guided task.	Child completes task with fewer hints and stable score.	Child completes task independently and improves score consistency.
Technology	Keyboard Key Match	Concept recall, interaction accuracy, speed, and retry improvement	Child identifies basic items and completes guided task.	Child completes task with fewer hints and stable score.	Child completes task independently and improves score consistency.
Technology	Internet Safety Quiz	Concept recall, interaction accuracy, speed, and retry improvement	Child identifies basic items and completes guided task.	Child completes task with fewer hints and stable score.	Child completes task independently and improves score consistency.
Technology	Binary Number Puzzle	Concept recall, interaction accuracy, speed, and retry improvement	Child identifies basic items and completes guided task.	Child completes task with fewer hints and stable score.	Child completes task independently and improves score consistency.

Subject	Game	Skill Area	Beginner Evidence	Intermediate Evidence	Advanced Evidence
Technology	AI vs Human Sorting	Concept recall, interaction accuracy, speed, and retry improvement	Child identifies basic items and completes guided task.	Child completes task with fewer hints and stable score.	Child completes task independently and improves score consistency.
Technology	Tech Timeline Puzzle	Concept recall, interaction accuracy, speed, and retry improvement	Child identifies basic items and completes guided task.	Child completes task with fewer hints and stable score.	Child completes task independently and improves score consistency.
Technology	Gadget Recognition Game	Concept recall, interaction accuracy, speed, and retry improvement	Child identifies basic items and completes guided task.	Child completes task with fewer hints and stable score.	Child completes task independently and improves score consistency.
English	Alphabet Matching	Concept recall, interaction accuracy, speed, and retry improvement	Child identifies basic items and completes guided task.	Child completes task with fewer hints and stable score.	Child completes task independently and improves score consistency.
English	Spelling Puzzle	Concept recall, interaction accuracy, speed, and retry improvement	Child identifies basic items and completes guided task.	Child completes task with fewer hints and stable score.	Child completes task independently and improves score consistency.
English	Word and Picture Match	Concept recall, interaction accuracy, speed, and retry improvement	Child identifies basic items and completes guided task.	Child completes task with fewer hints and stable score.	Child completes task independently and improves score consistency.
English	Fill in the Blanks	Concept recall, interaction accuracy, speed, and retry improvement	Child identifies basic items and completes guided task.	Child completes task with fewer hints and stable score.	Child completes task independently and improves score consistency.
English	Sentence Builder	Concept recall, interaction accuracy, speed, and retry improvement	Child identifies basic items and completes guided task.	Child completes task with fewer hints and stable score.	Child completes task independently and improves score consistency.
English	Rhyming Words	Concept recall, interaction accuracy, speed, and retry improvement	Child identifies basic items and completes guided task.	Child completes task with fewer hints and stable score.	Child completes task independently and improves score consistency.
English	Grammar Quiz	Concept recall, interaction accuracy, speed, and retry improvement	Child identifies basic items and completes guided task.	Child completes task with fewer hints and stable score.	Child completes task independently and improves score consistency.
English	Story Ordering	Concept recall, interaction accuracy, speed, and retry improvement	Child identifies basic items and completes guided task.	Child completes task with fewer hints and stable score.	Child completes task independently and improves score consistency.
English	Sight Word Memory	Concept recall, interaction accuracy, speed, and retry improvement	Child identifies basic items and completes guided task.	Child completes task with fewer hints and stable score.	Child completes task independently and improves score consistency.

G.1 Rubric Explanation

Each mini-game is evaluated with the same educational rubric so that scores are not treated as random entertainment results. A correct answer, match, drag operation, or quiz selection becomes evidence of learning. This allows the parent portal to convert game attempts into understandable progress indicators. The rubric also supports future expansion because any new game can be inserted into the same scoring structure without changing the whole reporting model.

G.2 Mathematics Learning Map

Table 134: Mathematics learning map

Game	Primary Concept	Feedback Type	Parent Report Meaning
Balloon Pop	Mathematics curriculum concept linked with balloon pop	Immediate visual feedback plus retry option	Shows practice frequency, score trend, and subject interest
Counting Fun	Mathematics curriculum concept linked with counting fun	Immediate visual feedback plus retry option	Shows practice frequency, score trend, and subject interest
Pattern Match	Mathematics curriculum concept linked with pattern match	Immediate visual feedback plus retry option	Shows practice frequency, score trend, and subject interest
Clock Reading	Mathematics curriculum concept linked with clock reading	Immediate visual feedback plus retry option	Shows practice frequency, score trend, and subject interest
Math Maze	Mathematics curriculum concept linked with math maze	Immediate visual feedback plus retry option	Shows practice frequency, score trend, and subject interest
Fraction Match	Mathematics curriculum concept linked with fraction match	Immediate visual feedback plus retry option	Shows practice frequency, score trend, and subject interest
Multiplication Quiz	Mathematics curriculum concept linked with multiplication quiz	Immediate visual feedback plus retry option	Shows practice frequency, score trend, and subject interest

G.2 Science Learning Map

Table 135: Science learning map

Game	Primary Concept	Feedback Type	Parent Report Meaning
Human Body Match	Science curriculum concept linked with human body match	Immediate visual feedback plus retry option	Shows practice frequency, score trend, and subject interest
Life Cycle Puzzle	Science curriculum concept linked with life cycle puzzle	Immediate visual feedback plus retry option	Shows practice frequency, score trend, and subject interest
Weather Guessing	Science curriculum concept linked with weather guessing	Immediate visual feedback plus retry option	Shows practice frequency, score trend, and subject interest
Animal Sounds and Habitats	Science curriculum concept linked with animal sounds and habitats	Immediate visual feedback plus retry option	Shows practice frequency, score trend, and subject interest
Plant Parts	Science curriculum concept linked with plant parts	Immediate visual feedback plus retry option	Shows practice frequency, score trend, and subject interest
Sink or Float	Science curriculum concept linked with sink or float	Immediate visual feedback plus retry option	Shows practice frequency, score trend, and subject interest
Water Cycle	Science curriculum concept linked with water cycle	Immediate visual feedback plus retry option	Shows practice frequency, score trend, and subject interest
Science Tools	Science curriculum concept linked with science tools	Immediate visual feedback plus retry option	Shows practice frequency, score trend, and subject interest
States of Matter	Science curriculum concept linked with states of matter	Immediate visual feedback plus retry option	Shows practice frequency, score trend, and subject interest

G.2 Technology Learning Map

Table 136: Technology learning map

Game	Primary Concept	Feedback Type	Parent Report Meaning
Computer Parts Match	Technology curriculum concept linked with computer parts match	Immediate visual feedback plus retry option	Shows practice frequency, score trend, and subject interest
Robot Path Puzzle	Technology curriculum concept linked with robot path puzzle	Immediate visual feedback plus retry option	Shows practice frequency, score trend, and subject interest
Keyboard Key Match	Technology curriculum concept linked with keyboard key match	Immediate visual feedback plus retry option	Shows practice frequency, score trend, and subject interest
Internet Safety Quiz	Technology curriculum concept linked with internet safety quiz	Immediate visual feedback plus retry option	Shows practice frequency, score trend, and subject interest
Binary Number Puzzle	Technology curriculum concept linked with binary number puzzle	Immediate visual feedback plus retry option	Shows practice frequency, score trend, and subject interest
AI vs Human Sorting	Technology curriculum concept linked with ai vs human sorting	Immediate visual feedback plus retry option	Shows practice frequency, score trend, and subject interest
Tech Timeline Puzzle	Technology curriculum concept linked with tech timeline puzzle	Immediate visual feedback plus retry option	Shows practice frequency, score trend, and subject interest
Gadget Recognition Game	Technology curriculum concept linked with gadget recognition game	Immediate visual feedback plus retry option	Shows practice frequency, score trend, and subject interest

G.2 English Learning Map

Table 137: English learning map

Game	Primary Concept	Feedback Type	Parent Report Meaning
Alphabet Matching	English curriculum concept linked with alphabet matching	Immediate visual feedback plus retry option	Shows practice frequency, score trend, and subject interest
Spelling Puzzle	English curriculum concept linked with spelling puzzle	Immediate visual feedback plus retry option	Shows practice frequency, score trend, and subject interest
Word and Picture Match	English curriculum concept linked with word and picture match	Immediate visual feedback plus retry option	Shows practice frequency, score trend, and subject interest
Fill in the Blanks	English curriculum concept linked with fill in the blanks	Immediate visual feedback plus retry option	Shows practice frequency, score trend, and subject interest
Sentence Builder	English curriculum concept linked with sentence builder	Immediate visual feedback plus retry option	Shows practice frequency, score trend, and subject interest
Rhyming Words	English curriculum concept linked with rhyming words	Immediate visual feedback plus retry option	Shows practice frequency, score trend, and subject interest
Grammar Quiz	English curriculum concept linked with grammar quiz	Immediate visual feedback plus retry option	Shows practice frequency, score trend, and subject interest
Story Ordering	English curriculum concept linked with story ordering	Immediate visual feedback plus retry option	Shows practice frequency, score trend, and subject interest

Game	Primary Concept	Feedback Type	Parent Report Meaning
Sight Word Memory	English curriculum concept linked with sight word memory	Immediate visual feedback plus retry option	Shows practice frequency, score trend, and subject interest

Appendix H. Complete User Stories and Acceptance Criteria

Table 138: User stories and acceptance criteria

ID	Role	User Story	Acceptance Criteria	Priority
US-P-1	Parent	As a parent, I want to perform Gamefide task 1 so that the learning workflow remains complete.	Feature works with valid input and gives clear feedback for invalid input.	High
US-P-2	Parent	As a parent, I want to perform Gamefide task 2 so that the learning workflow remains complete.	Feature works with valid input and gives clear feedback for invalid input.	High
US-P-3	Parent	As a parent, I want to perform Gamefide task 3 so that the learning workflow remains complete.	Feature works with valid input and gives clear feedback for invalid input.	High
US-P-4	Parent	As a parent, I want to perform Gamefide task 4 so that the learning workflow remains complete.	Feature works with valid input and gives clear feedback for invalid input.	High
US-P-5	Parent	As a parent, I want to perform Gamefide task 5 so that the learning workflow remains complete.	Feature works with valid input and gives clear feedback for invalid input.	High
US-P-6	Parent	As a parent, I want to perform Gamefide task 6 so that the learning workflow remains complete.	Feature works with valid input and gives clear feedback for invalid input.	High
US-P-7	Parent	As a parent, I want to perform Gamefide task 7 so that the learning workflow remains complete.	Feature works with valid input and gives clear feedback for invalid input.	High
US-P-8	Parent	As a parent, I want to perform Gamefide task 8 so that the learning workflow remains complete.	Feature works with valid input and gives clear feedback for invalid input.	High
US-P-9	Parent	As a parent, I want to perform Gamefide task 9 so that the learning workflow remains complete.	Feature works with valid input and gives clear feedback for invalid input.	Medium
US-P-10	Parent	As a parent, I want to perform Gamefide task 10 so that the learning workflow remains complete.	Feature works with valid input and gives clear feedback for invalid input.	Medium
US-P-11	Parent	As a parent, I want to perform Gamefide task 11 so that the learning workflow remains complete.	Feature works with valid input and gives clear feedback for invalid input.	Medium
US-P-12	Parent	As a parent, I want to perform Gamefide task 12 so that the learning workflow remains complete.	Feature works with valid input and gives clear feedback for invalid input.	Medium
US-P-13	Parent	As a parent, I want to perform Gamefide task 13 so that the learning workflow remains complete.	Feature works with valid input and gives clear feedback for invalid input.	Medium
US-P-14	Parent	As a parent, I want to perform Gamefide task 14 so that the learning workflow remains complete.	Feature works with valid input and gives clear feedback for invalid input.	Medium
US-P-15	Parent	As a parent, I want to perform Gamefide task 15 so that the learning workflow remains complete.	Feature works with valid input and gives clear feedback for invalid input.	Medium
US-C-1	Child	As a child, I want to perform Gamefide task 1 so that the learning workflow remains complete.	Feature works with valid input and gives clear feedback for invalid input.	High
US-C-2	Child	As a child, I want to perform Gamefide task 2 so that the learning workflow remains complete.	Feature works with valid input and gives clear feedback for invalid input.	High
US-C-3	Child	As a child, I want to perform Gamefide task 3 so that the learning workflow remains complete.	Feature works with valid input and gives clear feedback for invalid input.	High
US-C-4	Child	As a child, I want to perform Gamefide task 4 so that the learning workflow remains complete.	Feature works with valid input and gives clear feedback for invalid input.	High
US-C-5	Child	As a child, I want to perform Gamefide task 5 so that the learning workflow remains complete.	Feature works with valid input and gives clear feedback for invalid input.	High
US-C-6	Child	As a child, I want to perform Gamefide task 6 so that the learning workflow remains complete.	Feature works with valid input and gives clear feedback for invalid input.	High
US-C-7	Child	As a child, I want to perform Gamefide task 7 so that the learning workflow remains complete.	Feature works with valid input and gives clear feedback for invalid input.	High
US-C-8	Child	As a child, I want to perform Gamefide task 8 so that the learning workflow remains complete.	Feature works with valid input and gives clear feedback for invalid input.	High
US-C-9	Child	As a child, I want to perform Gamefide task 9 so that the learning workflow remains complete.	Feature works with valid input and gives clear feedback for invalid input.	Medium
US-C-10	Child	As a child, I want to perform Gamefide task 10 so that the learning workflow remains complete.	Feature works with valid input and gives clear feedback for invalid input.	Medium
US-C-11	Child	As a child, I want to perform Gamefide task 11 so that the learning workflow remains complete.	Feature works with valid input and gives clear feedback for invalid input.	Medium
US-C-12	Child	As a child, I want to perform Gamefide task 12 so that the learning workflow remains complete.	Feature works with valid input and gives clear feedback for invalid input.	Medium
US-C-13	Child	As a child, I want to perform Gamefide task 13 so that the learning workflow remains complete.	Feature works with valid input and gives clear feedback for invalid input.	Medium
US-C-14	Child	As a child, I want to perform Gamefide task 14 so that the learning workflow remains complete.	Feature works with valid input and gives clear feedback for invalid input.	Medium

ID	Role	User Story	Acceptance Criteria	Priority
US-C-15	Child	As a child, I want to perform Gamefide task 15 so that the learning workflow remains complete.	Feature works with valid input and gives clear feedback for invalid input.	Medium
US-A-1	Administrator	As a administrator, I want to perform Gamefide task 1 so that the learning workflow remains complete.	Feature works with valid input and gives clear feedback for invalid input.	High
US-A-2	Administrator	As a administrator, I want to perform Gamefide task 2 so that the learning workflow remains complete.	Feature works with valid input and gives clear feedback for invalid input.	High
US-A-3	Administrator	As a administrator, I want to perform Gamefide task 3 so that the learning workflow remains complete.	Feature works with valid input and gives clear feedback for invalid input.	High
US-A-4	Administrator	As a administrator, I want to perform Gamefide task 4 so that the learning workflow remains complete.	Feature works with valid input and gives clear feedback for invalid input.	High
US-A-5	Administrator	As a administrator, I want to perform Gamefide task 5 so that the learning workflow remains complete.	Feature works with valid input and gives clear feedback for invalid input.	High
US-A-6	Administrator	As a administrator, I want to perform Gamefide task 6 so that the learning workflow remains complete.	Feature works with valid input and gives clear feedback for invalid input.	High
US-A-7	Administrator	As a administrator, I want to perform Gamefide task 7 so that the learning workflow remains complete.	Feature works with valid input and gives clear feedback for invalid input.	High
US-A-8	Administrator	As a administrator, I want to perform Gamefide task 8 so that the learning workflow remains complete.	Feature works with valid input and gives clear feedback for invalid input.	High
US-A-9	Administrator	As a administrator, I want to perform Gamefide task 9 so that the learning workflow remains complete.	Feature works with valid input and gives clear feedback for invalid input.	Medium
US-A-10	Administrator	As a administrator, I want to perform Gamefide task 10 so that the learning workflow remains complete.	Feature works with valid input and gives clear feedback for invalid input.	Medium
US-A-11	Administrator	As a administrator, I want to perform Gamefide task 11 so that the learning workflow remains complete.	Feature works with valid input and gives clear feedback for invalid input.	Medium
US-A-12	Administrator	As a administrator, I want to perform Gamefide task 12 so that the learning workflow remains complete.	Feature works with valid input and gives clear feedback for invalid input.	Medium
US-A-13	Administrator	As a administrator, I want to perform Gamefide task 13 so that the learning workflow remains complete.	Feature works with valid input and gives clear feedback for invalid input.	Medium
US-A-14	Administrator	As a administrator, I want to perform Gamefide task 14 so that the learning workflow remains complete.	Feature works with valid input and gives clear feedback for invalid input.	Medium
US-A-15	Administrator	As a administrator, I want to perform Gamefide task 15 so that the learning workflow remains complete.	Feature works with valid input and gives clear feedback for invalid input.	Medium

H.1 Story Prioritization

Parent and child stories were prioritized first because they represent the main educational workflow. Administrative stories are also important, but the prototype mainly focuses on the child learning experience and parent monitoring. Acceptance criteria emphasize visible UI feedback, secure data storage, and measurable progress updates.

Appendix I. Detailed API Specification Matrix

Table 139: Detailed API specification matrix

Route	Method	Auth	Request Data	Success Response	Validation / Error Handling
/api/auth/register	POST	JWT where protected	JSON body, route params, or form-data	Status, data object, message, or updated record	Joi validation, JWT middleware, Mongoose error handling, safe message
/api/auth/login-parent	POST	JWT where protected	JSON body, route params, or form-data	Status, data object, message, or updated record	Joi validation, JWT middleware, Mongoose error handling, safe message
/api/auth/register-child	POST	JWT where protected	JSON body, route params, or form-data	Status, data object, message, or updated record	Joi validation, JWT middleware, Mongoose error handling, safe message
/api/auth/child-login	POST	JWT where protected	JSON body, route params, or form-data	Status, data object, message, or updated record	Joi validation, JWT middleware, Mongoose error handling, safe message
/api/parent/dashboard	GET	JWT where protected	JSON body, route params, or form-data	Status, data object, message, or updated record	Joi validation, JWT middleware, Mongoose error handling, safe message
/api/parent/reports/:childId	GET	JWT where protected	JSON body, route params, or form-data	Status, data object, message, or updated record	Joi validation, JWT middleware, Mongoose error handling, safe message
/api/users/children	GET	JWT where protected	JSON body, route params, or form-data	Status, data object, message, or updated record	Joi validation, JWT middleware, Mongoose error handling, safe message
/api/progress	POST	JWT where protected	JSON body, route params, or form-data	Status, data object, message, or updated record	Joi validation, JWT middleware, Mongoose error handling, safe message
/api/achievements	POST	JWT where protected	JSON body, route params, or form-data	Status, data object, message, or updated record	Joi validation, JWT middleware, Mongoose error handling, safe message
/api/quiz/:subject	GET	JWT where protected	JSON body, route params, or form-data	Status, data object, message, or updated record	Joi validation, JWT middleware, Mongoose error handling, safe message
/api/quiz/submit	POST	JWT where protected	JSON body, route params, or form-data	Status, data object, message, or updated record	Joi validation, JWT middleware, Mongoose error handling, safe message
/api/game/start	POST	JWT where protected	JSON body, route params, or form-data	Status, data object, message, or updated record	Joi validation, JWT middleware, Mongoose error handling, safe message
/api/game/submit	POST	JWT where protected	JSON body, route params, or form-data	Status, data object, message, or updated record	Joi validation, JWT middleware, Mongoose error handling, safe message
/api/stories/generate	POST	JWT where protected	JSON body, route params, or form-data	Status, data object, message, or updated record	Joi validation, JWT middleware, Mongoose error handling, safe message

I.1 API Design Notes

- The API uses route grouping so authentication, parent, users, progress, achievements, quiz, game, and story responsibilities remain separate.

- Protected routes require JWT verification before controller logic executes.

- Form-data is used where avatar upload is required.

- Story generation route sends controlled prompt information to the AI integration rather than accepting unrestricted prompts.

- Swagger UI documents routes for testing and developer review.

Appendix J. Security, Privacy, and Validation Matrix

Table 140: Security and validation matrix

Area	Risk	Control Used	Verification Method	Future Enhancement
Parent login	Unauthorized access	JWT and password/social authentication	Negative login tests	Rate limiting
Child PIN	Weak child credential	Bcrypt hashed PIN and simple UI	Wrong PIN test	PIN reset flow
Biometric login	Device identity mismatch	Native biometric plugin and sparse fingerprint field	Device login test	Device binding audit
API payload	Invalid data	Joi validation and Mongoose schema constraints	Invalid age/score tests	Central validation schema
Database	Duplicate records	Unique indexes and references	Duplicate email/progress test	Migration scripts
AI story	Uncontrolled text	Prompt templates and story type constraints	Generate three story types	Moderation layer
Reports	Wrong child access	Parent-child ownership checking	Wrong parent access test	Audit logs
Headers/CORS	Browser security	Helmet and configured CORS	Route smoke tests	Production domain allowlist

J.1 Child Privacy Considerations

Gamefide stores only the data needed for the learning workflow, such as child name, avatar reference, age, score, level, progress, and generated stories. Passwords and PIN values should remain hashed, and parent-child ownership must be enforced by backend authorization checks. Because the application targets children aged 7 to 12, future production deployment should add parental consent wording, clear data deletion options, and stronger production logging controls.

Appendix K. Screen-by-Screen HCI Checklist

Table 141: HCI checklist for all implemented screens

Screen	Visibility	Simplicity	Error Prevention	Feedback	Accessibility Note
Splash / Welcome Page	Important actions are visible through large cards/buttons.	Text is short and child-friendly where needed.	Forms should validate before submission.	Screen shows route change, dialog, chart, or result.	Future audio narration and larger contrast options can improve inclusion.
Parent Login Page	Important actions are visible through large cards/buttons.	Text is short and child-friendly where needed.	Forms should validate before submission.	Screen shows route change, dialog, chart, or result.	Future audio narration and larger contrast options can improve inclusion.
Parent Dashboard Empty State	Important actions are visible through large cards/buttons.	Text is short and child-friendly where needed.	Forms should validate before submission.	Screen shows route change, dialog, chart, or result.	Future audio narration and larger contrast options can improve inclusion.
Add Child Page	Important actions are visible through large cards/buttons.	Text is short and child-friendly where needed.	Forms should validate before submission.	Screen shows route change, dialog, chart, or result.	Future audio narration and larger contrast options can improve inclusion.
Parent Dashboard with Child	Important actions are visible through large cards/buttons.	Text is short and child-friendly where needed.	Forms should validate before submission.	Screen shows route change, dialog, chart, or result.	Future audio narration and larger contrast options can improve inclusion.
Parent Reports Child List	Important actions are visible through large cards/buttons.	Text is short and child-friendly where needed.	Forms should validate before submission.	Screen shows route change, dialog, chart, or result.	Future audio narration and larger contrast options can improve inclusion.
Parent Profile Page	Important actions are visible through large cards/buttons.	Text is short and child-friendly where needed.	Forms should validate before submission.	Screen shows route change, dialog, chart, or result.	Future audio narration and larger contrast options can improve inclusion.
Logout Confirmation Dialog	Important actions are visible through large cards/buttons.	Text is short and child-friendly where needed.	Forms should validate before submission.	Screen shows route change, dialog, chart, or result.	Future audio narration and larger contrast options can improve inclusion.
Role Selection / Child Login Entry	Important actions are visible through large cards/buttons.	Text is short and child-friendly where needed.	Forms should validate before submission.	Screen shows route change, dialog, chart, or result.	Future audio narration and larger contrast options can improve inclusion.
Child PIN Login Dialog	Important actions are visible through large cards/buttons.	Text is short and child-friendly where needed.	Forms should validate before submission.	Screen shows route change, dialog, chart, or result.	Future audio narration and larger contrast options can improve inclusion.
Child Home Dashboard	Important actions are visible through large cards/buttons.	Text is short and child-friendly where needed.	Forms should validate before submission.	Screen shows route change, dialog, chart, or result.	Future audio narration and larger contrast options can improve inclusion.
Games Home Page	Important actions are visible through large cards/buttons.	Text is short and child-friendly where needed.	Forms should validate before submission.	Screen shows route change, dialog, chart, or result.	Future audio narration and larger contrast options can improve inclusion.
English Games Extended List	Important actions are visible through large cards/buttons.	Text is short and child-friendly where needed.	Forms should validate before submission.	Screen shows route change, dialog, chart, or result.	Future audio narration and larger contrast options can improve inclusion.
English Learning Games Page	Important actions are visible through large cards/buttons.	Text is short and child-friendly where needed.	Forms should validate before submission.	Screen shows route change, dialog, chart, or result.	Future audio narration and larger contrast options can improve inclusion.
Math Games Page	Important actions are visible through large cards/buttons.	Text is short and child-friendly where needed.	Forms should validate before submission.	Screen shows route change, dialog, chart, or result.	Future audio narration and larger contrast options can improve inclusion.
Math Maze Game Screen	Important actions are visible through large cards/buttons.	Text is short and child-friendly where needed.	Forms should validate before submission.	Screen shows route change, dialog, chart, or result.	Future audio narration and larger contrast options can improve inclusion.
Science Games Page	Important actions are visible through large cards/buttons.	Text is short and child-friendly where needed.	Forms should validate before submission.	Screen shows route change, dialog, chart, or result.	Future audio narration and larger contrast options can improve inclusion.
Technology Games Page	Important actions are visible through large cards/buttons.	Text is short and child-friendly where needed.	Forms should validate before submission.	Screen shows route change, dialog, chart, or result.	Future audio narration and larger contrast options can improve inclusion.
Subjects Page	Important actions are visible through large cards/buttons.	Text is short and child-friendly where needed.	Forms should validate before submission.	Screen shows route change, dialog, chart, or result.	Future audio narration and larger contrast options can improve inclusion.
40-Level Quiz Selector	Important actions are visible through large cards/buttons.	Text is short and child-friendly where needed.	Forms should validate before submission.	Screen shows route change, dialog, chart, or result.	Future audio narration and larger contrast options can improve inclusion.
AI Story Magic Page	Important actions are visible through large cards/buttons.	Text is short and child-friendly where needed.	Forms should validate before submission.	Screen shows route change, dialog, chart, or result.	Future audio narration and larger contrast options can improve inclusion.

Screen	Visibility	Simplicity	Error Prevention	Feedback	Accessibility Note
Child Profile and Achievements Page	Important actions are visible through large cards/buttons.	Text is short and child-friendly where needed.	Forms should validate before submission.	Screen shows route change, dialog, chart, or result.	Future audio narration and larger contrast options can improve inclusion.

K.1 HCI Design Discussion

The interface uses mobile-first pages, colorful cards, rounded sections, visual hierarchy, and role-specific navigation. Child pages favor game cards and quick feedback, while parent pages prioritize management, reports, and child account settings. The design avoids heavy forms for children and keeps parent controls in the guardian portal.

Appendix L. Team Contribution and Work Breakdown Structure

Table 142: Five-member team contribution table

Member	Primary Responsibility	Major Deliverables	Verification Evidence
Student 1	Frontend UI	Ionic screens, routing, styling	Screens, diagrams, tables, and test cases included in report
Student 2	Backend API	Express controllers, route middleware	Screens, diagrams, tables, and test cases included in report
Student 3	Database and models	Mongoose schemas, indexes, relations	Screens, diagrams, tables, and test cases included in report
Student 4	Game implementation	Mini-games, quiz flows, scoring	Screens, diagrams, tables, and test cases included in report
Student 5	Testing and documentation	Test cases, screenshots, final report	Screens, diagrams, tables, and test cases included in report

Table 143: Work breakdown structure

WBS Code	Work Package	Output	Status
WBS-1	Requirement analysis	Requirement analysis deliverable completed	Complete
WBS-2	Template study	Template study deliverable completed	Complete
WBS-3	UI design	UI design deliverable completed	Complete
WBS-4	Authentication	Authentication deliverable completed	Complete
WBS-5	Parent portal	Parent portal deliverable completed	Complete
WBS-6	Child portal	Child portal deliverable completed	Complete
WBS-7	Games catalog	Games catalog deliverable completed	Complete
WBS-8	Quiz engine	Quiz engine deliverable completed	Complete
WBS-9	AI Storybook	AI Storybook deliverable completed	Complete
WBS-10	Progress module	Progress module deliverable completed	Complete
WBS-11	Reports module	Reports module deliverable completed	Complete
WBS-12	Database design	Database design deliverable completed	Complete
WBS-13	API documentation	API documentation deliverable completed	Complete
WBS-14	Testing	Testing deliverable completed	Complete
WBS-15	Final report	Final report deliverable completed	Complete

Appendix M. Maintenance, Deployment, and Troubleshooting Plan

Table 144: Maintenance and troubleshooting plan

Issue	Possible Cause	Resolution	Responsible Module
Frontend does not start	Node modules missing	Run npm install and ionic serve	Frontend
API cannot connect	MongoDB URI wrong	Check .env MONGO_URI and database service	Backend
Story not generated	Gemini key missing or network error	Check GEMINI_API_KEY and API response	AI Story
Protected route fails	JWT expired or missing	Login again or refresh token	Authentication
Avatar upload fails	Multipart form mismatch	Check field name and storage directory	Child profile
Score not saved	Session ID or child ID invalid	Validate request payload and database refs	Game
Chart empty	No progress records	Play games or seed progress data	Reports
Level remains locked	Previous level score below threshold	Score at least 3/5 in prior level	Quiz

M.1 Production Readiness Checklist

Table 145: Production readiness checklist

Checklist Item	Status in Prototype	Production Action
Environment variables	Configured locally	Use secret manager
HTTPS	Assumed for deployment	Enforce HTTPS only
Database backups	Not part of prototype	Schedule automated backups
Monitoring	Manual testing	Add request/error logs
Rate limiting	Suggested	Add express-rate-limit
AI safety	Prompt templates	Add moderation and abuse limits
App packaging	Capacitor supported	Sign Android/iOS builds
Accessibility	Visual UI complete	Add narration and contrast testing

Appendix N. Future Enhancements and Research Direction

Table 146: Future enhancement roadmap

Enhancement	Reason	Expected Benefit	Complexity
Teacher dashboard	Improves learning, monitoring, access, or scalability	Higher engagement and stronger educational value	Medium
Adaptive difficulty engine	Improves learning, monitoring, access, or scalability	Higher engagement and stronger educational value	Medium
Voice narration for early readers	Improves learning, monitoring, access, or scalability	Higher engagement and stronger educational value	Medium
Offline mode for selected games	Improves learning, monitoring, access, or scalability	Higher engagement and stronger educational value	Medium
Saved story library	Improves learning, monitoring, access, or scalability	Higher engagement and stronger educational value	Medium
Multilingual Urdu/English content	Improves learning, monitoring, access, or scalability	Higher engagement and stronger educational value	Medium
Parent notification alerts	Improves learning, monitoring, access, or scalability	Higher engagement and stronger educational value	Medium
Leaderboard with privacy controls	Improves learning, monitoring, access, or scalability	Higher engagement and stronger educational value	Medium
AI content moderation	Improves learning, monitoring, access, or scalability	Higher engagement and stronger educational value	Medium
More STEM simulations	Improves learning, monitoring, access, or scalability	Higher engagement and stronger educational value	Medium
Learning recommendation system	Improves learning, monitoring, access, or scalability	Higher engagement and stronger educational value	Medium
Exportable PDF reports	Improves learning, monitoring, access, or scalability	Higher engagement and stronger educational value	Medium
Classroom mode	Improves learning, monitoring, access, or scalability	Higher engagement and stronger educational value	Medium
Accessibility settings	Improves learning, monitoring, access, or scalability	Higher engagement and stronger educational value	Medium
App store publishing	Improves learning, monitoring, access, or scalability	Higher engagement and stronger educational value	Medium

N.1 Research Extension

The project can be extended into research on whether gamified STEM learning improves motivation and retention for children aged 7 to 12. Future evaluation may compare quiz scores before and after repeated game sessions, measure parent engagement through report views, and analyze which game mechanics lead to better learning streaks.

Appendix O. Extended Traceability Matrix

Table 147: Extended requirements traceability matrix

Requirement	Linked Use Case	Linked Module	Linked Test	Evidence
FR-1: Register Parent Account	UC-1	Parent	TC-1	Implemented through route, screen, model, and test documentation
FR-2: Login Parent	UC-2	Child	TC-2	Implemented through route, screen, model, and test documentation
FR-3: Register Child Profile	UC-3	Game	TC-3	Implemented through route, screen, model, and test documentation
FR-4: Register Child with Fingerprint	UC-4	Quiz	TC-4	Implemented through route, screen, model, and test documentation
FR-5: Child Login	UC-5	Story	TC-5	Implemented through route, screen, model, and test documentation
FR-6: View Child Home Dashboard	UC-6	Report	TC-6	Implemented through route, screen, model, and test documentation
FR-7: Select Subject	UC-7	Achievement	TC-7	Implemented through route, screen, model, and test documentation
FR-8: Play Educational Game	UC-8	Auth	TC-8	Implemented through route, screen, model, and test documentation
FR-9: Submit Game Score	UC-9	Parent	TC-9	Implemented through route, screen, model, and test documentation
FR-10: Attempt Level Quiz	UC-10	Child	TC-10	Implemented through route, screen, model, and test documentation
FR-11: Generate AI Story	UC-11	Game	TC-11	Implemented through route, screen, model, and test documentation
FR-12: View Achievements	UC-12	Quiz	TC-12	Implemented through route, screen, model, and test documentation
FR-13: View Parent Dashboard	UC-13	Story	TC-13	Implemented through route, screen, model, and test documentation
FR-14: View Reports	UC-14	Report	TC-14	Implemented through route, screen, model, and test documentation
FR-15: Manage Quiz Questions	UC-15	Achievement	TC-15	Implemented through route, screen, model, and test documentation

O.1 Traceability Notes

Traceability connects stakeholder needs with implemented modules and tests. It shows that Gamefide is not only a collection of screens, but a structured software system where each requirement maps to UI actions, backend routes, data models, and verification cases.

Appendix P. Expanded Database Collection Notes

P.1 User Collection Notes

The User collection/model is part of the MongoDB persistence design. It supports the system by storing normalized or semi-structured data needed by the parent portal, child portal, games, achievements, quizzes, and AI story workflow. In Mongoose, schema-level constraints define required fields, default values, enums, references, and timestamps where needed.

Table 148: User collection notes

Aspect	Description
Purpose	Supports User-related data storage and retrieval.
Important Fields	email, password hash, googleId, appleId, userType, refreshToken
Validation	Required fields, enum values, unique constraints, and reference integrity where applicable.
Common Operations	Create, read, update, delete, query by child, query by parent, or aggregate score.
Reporting Impact	Feeds dashboard, progress charts, achievements, quiz state, or story history.

P.2 Parent Collection Notes

The Parent collection/model is part of the MongoDB persistence design. It supports the system by storing normalized or semi-structured data needed by the parent portal, child portal, games, achievements, quizzes, and AI story workflow. In Mongoose, schema-level constraints define required fields, default values, enums, references, and timestamps where needed.

Table 149: Parent collection notes

Aspect	Description
Purpose	Supports Parent-related data storage and retrieval.
Important Fields	user reference, children references
Validation	Required fields, enum values, unique constraints, and reference integrity where applicable.
Common Operations	Create, read, update, delete, query by child, query by parent, or aggregate score.
Reporting Impact	Feeds dashboard, progress charts, achievements, quiz state, or story history.

P.3 Child Collection Notes

The Child collection/model is part of the MongoDB persistence design. It supports the system by storing normalized or semi-structured data needed by the parent portal, child portal, games, achievements, quizzes, and AI story workflow. In Mongoose, schema-level constraints define required fields, default values, enums, references, and timestamps where needed.

Table 150: Child collection notes

Aspect	Description
Purpose	Supports Child-related data storage and retrieval.
Important Fields	parentId, name, avatar, fingerprint, pin hash, age, points, level, settings
Validation	Required fields, enum values, unique constraints, and reference integrity where applicable.
Common Operations	Create, read, update, delete, query by child, query by parent, or aggregate score.
Reporting Impact	Feeds dashboard, progress charts, achievements, quiz state, or story history.

P.4 Progress Collection Notes

The Progress collection/model is part of the MongoDB persistence design. It supports the system by storing normalized or semi-structured data needed by the parent portal, child portal, games, achievements, quizzes, and AI story workflow. In Mongoose, schema-level constraints define required fields, default values, enums, references, and timestamps where needed.

Table 151: Progress collection notes

Aspect	Description
Purpose	Supports Progress-related data storage and retrieval.
Important Fields	childId, subject, score, level, unique child-subject index
Validation	Required fields, enum values, unique constraints, and reference integrity where applicable.
Common Operations	Create, read, update, delete, query by child, query by parent, or aggregate score.
Reporting Impact	Feeds dashboard, progress charts, achievements, quiz state, or story history.

P.5 Achievement Collection Notes

The Achievement collection/model is part of the MongoDB persistence design. It supports the system by storing normalized or semi-structured data needed by the parent portal, child portal, games, achievements, quizzes, and AI story workflow. In Mongoose, schema-level constraints define required fields, default values, enums, references, and timestamps where needed.

Table 152: Achievement collection notes

Aspect	Description
Purpose	Supports Achievement-related data storage and retrieval.
Important Fields	childId, total_score, subject score arrays
Validation	Required fields, enum values, unique constraints, and reference integrity where applicable.
Common Operations	Create, read, update, delete, query by child, query by parent, or aggregate score.
Reporting Impact	Feeds dashboard, progress charts, achievements, quiz state, or story history.

P.6 TotalAchievement Collection Notes

The TotalAchievement collection/model is part of the MongoDB persistence design. It supports the system by storing normalized or semi-structured data needed by the parent portal, child portal, games, achievements, quizzes, and AI story workflow. In Mongoose, schema-level constraints define required fields, default values, enums, references, and timestamps where needed.

Table 153: TotalAchievement collection notes

Aspect	Description
Purpose	Supports TotalAchievement-related data storage and retrieval.
Important Fields	achievementId, total_achievement_score
Validation	Required fields, enum values, unique constraints, and reference integrity where applicable.
Common Operations	Create, read, update, delete, query by child, query by parent, or aggregate score.
Reporting Impact	Feeds dashboard, progress charts, achievements, quiz state, or story history.

P.7 Quiz Collection Notes

The Quiz collection/model is part of the MongoDB persistence design. It supports the system by storing normalized or semi-structured data needed by the parent portal, child portal, games, achievements, quizzes, and AI story workflow. In Mongoose, schema-level constraints define required fields, default values, enums, references, and timestamps where needed.

Table 154: Quiz collection notes

Aspect	Description
Purpose	Supports Quiz-related data storage and retrieval.
Important Fields	subject, questions, options, correctAnswer, difficulty
Validation	Required fields, enum values, unique constraints, and reference integrity where applicable.
Common Operations	Create, read, update, delete, query by child, query by parent, or aggregate score.
Reporting Impact	Feeds dashboard, progress charts, achievements, quiz state, or story history.

P.8 STEMContent Collection Notes

The STEMContent collection/model is part of the MongoDB persistence design. It supports the system by storing normalized or semi-structured data needed by the parent portal, child portal, games, achievements, quizzes, and AI story workflow. In Mongoose, schema-level constraints define required fields, default values, enums, references, and timestamps where needed.

Table 155: STEMContent collection notes

Aspect	Description
Purpose	Supports STEMContent-related data storage and retrieval.
Important Fields	title, description, subject, difficulty, contentType, objectives
Validation	Required fields, enum values, unique constraints, and reference integrity where applicable.
Common Operations	Create, read, update, delete, query by child, query by parent, or aggregate score.
Reporting Impact	Feeds dashboard, progress charts, achievements, quiz state, or story history.

P.9 Story Collection Notes

The Story collection/model is part of the MongoDB persistence design. It supports the system by storing normalized or semi-structured data needed by the parent portal, child portal, games, achievements, quizzes, and AI story workflow. In Mongoose, schema-level constraints define required fields, defaults, enums, references, and timestamps where needed.

Table 156: Story collection notes

Aspect	Description
Purpose	Supports Story-related data storage and retrieval.
Important Fields	childName, animal, storyType, generated content
Validation	Required fields, enum values, unique constraints, and reference integrity where applicable.
Common Operations	Create, read, update, delete, query by child, query by parent, or aggregate score.
Reporting Impact	Feeds dashboard, progress charts, achievements, quiz state, or story history.

Appendix Q. Final Submission Checklist

This checklist was prepared to show that the final PDF contains the major documentation artifacts normally expected in a BSCS final year project submission. It also helps the group review the document before printing, binding, or submitting it to the department.

Table 157: Final submission checklist

Checklist Area	Included in Report	Verification Note
University-style title page	Yes	Uses BSCS final project cover with University of Okara logo.
Technology chapters	Yes	Frontend, backend, database, AI, security, and deployment technologies are explained.
Template chapter order	Yes	The report follows introduction, overall description, requirements, architecture, design, testing, and conclusion structure.
ER diagram	Yes	MongoDB/Mongoose entities and relationships are included.
UML diagram	Yes	Controllers, models, and service collaboration are represented.
Sequence diagrams	Yes	Child registration, score submission, story generation, and reports are modeled.
Game sub-chapters	Yes	Math, Science, Technology, and English games are documented separately.
Screenshots	Yes	All supplied interface screenshots are inserted with explanation.
Testing tables	Yes	Functional test cases and summary of results are included.
Appendices	Yes	Glossary, IV&V, deployment, user manual, API examples, and extended matrices are included.
Minimum page target	Yes	The final generated document exceeds 80 pages.
Editable placeholders	Yes	Advisor name, group ID, and student details are intentionally left as placeholders for final team editing.

The final report is ready for review after replacing placeholders such as advisor name, group ID, student registration numbers, and student names. The technical and functional content is project-specific to Gamefide and can be used as the base version for university submission.